System Integration for Photonic Circuits:

A Tri-Channel Framework for Error, Loss, and Thermal Connectivity

From Component Certification through Correlated Disorder to System-Level Yield

Version History:
- **v0 (Mar 4, 2026):** Early LinkedIn post: "On My Upcoming Physics Colloquium Talk at NCU (March 4, 2026)"; thoughts on worked examples from basic to advanced
- **v7 (Mar 30, 2026):** Preliminary technical document.

Abstract

A photonic integrated circuit that passes every per-component specification may still fail at the system level — not because any single device is defective, but because three qualitatively different degradation channels interact in ways that component-level analysis cannot predict.

The central problem is *system-level yield*: given an N-component PIC fabricated on a real foundry process, what fraction of dies will meet specification, what is the power cost of thermal control, and does quantum operation provide a yield advantage? This problem is urgent as driven by markets. Practically we need a tool addresses fabrication yield, quantum fidelity, and thermal power budget at the system level.

This document introduces the **tri-channel (ELC) framework**, which decomposes system-level degradation into three channels with distinct physical origins, combination rules, and engineering currencies:

- **Channel 1 (Error):** Accumulated phase imprecision from fabrication disorder. Additive (RSS). Currency: fabrication precision (nm).
- **Channel 2 (Loss):** Irreversible quantum resource consumption. Multiplicative (product of fidelities). Currency: quantum fidelity.
- **Channel 3 (Connectivity):** Thermal mode-path matching — whether the intended circuit topology physically exists. Threshold character. Currency: **watts** of heater power.

Channel 3 is logically prior: one must first establish that mode-path connectivity is maintained before evaluating loss or error along that path. This hierarchy mirrors B4's resource hierarchy (Unitarity $\supset$ Coherence $\supset$ Entanglement) at the system level, and all three channels unify within a single path-integral action: phase $\rightarrow$ Ch 1, amplitude $\rightarrow$ Ch 2, saddle-point existence $\rightarrow$ Ch 3.

The framework is implemented through five modules: M1 (product-of-unitaries system simulator with 10^2–$10^3\times$ PI speedup over Monte Carlo), M2 (thermal connectivity engine with 2D Green's function, cross-talk matrix, and mode-overlap computation), M3 (self-consistent correlated tolerance model coupling fabrication and thermal disorder), M4 (tri-channel yield computation with binding-channel identification and tolerance allocation), and M5 (bridge alarm cascade for root-cause diagnostics). Each module is grounded in current literature and positioned within a three-tier perspective: T1 (traditional/industry standard), T2 (C6's contribution), T3 (novel architectures that eliminate the binding constraint).

Three test cases spanning the classical–quantum spectrum validate the framework:

- **TC1: 800G Transceiver** (20 components, O-band). Channel 3 binding: thermal tuning of ring MUX elements dominates the co-packaged power envelope. Corrective path: athermal ring design or substrate undercut (5–40$\times$ efficiency gain).
- **TC2: QKD Module** (10 components, C-band). Channel 2 binding: every decibel of chip insertion loss degrades the secure key rate. Corrective path: hybrid InP/SiN or TFLN for sub-2 dB chip loss.

- **TC3: Quantum Photonic Processor** (48 components, 8-mode Clements mesh). All three channels compete: fabrication error degrades unitary fidelity (Ch 1), photon loss limits survival (Ch 2), and 32 heaters strain the thermal budget (Ch 3). Corrective path: time-multiplexed GKP architecture.

Key quantitative findings: correlated fabrication disorder changes yield prediction by 5–15 percentage points versus independent models; the quantum yield advantage ΔY persists in the full tri-channel framework with crossover $N^* \approx 12$–20 for SOI; thermal tuning power for 16-channel WDM can consume 30–50% of the co-packaged optics power budget; and for $Y_{\text{sys}} \geq 90\%$ at $N = 40$ components, fabrication roughness must be held below $\sigma_w < 2$ nm — tighter than current foundry processes.

The tri-channel framework extends C4's dual-channel (Error vs. Loss) to the system level by adding a third, logically prior channel (Connectivity) whose currency is watts, not precision or fidelity. As part of Series C (production-grade toolkits for modern optical engineering), C6 provides the system integration infrastructure that follows component-level tolerance analysis (C2), design optimization (C3), and device certification (C4).

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List of Abbreviations

General

BER – Bit error rate
BOX – Buried oxide layer
DWDM – Dense WDM
ELC – Error–Loss–Connectivity
FOM – Figure of merit
GC – Grating coupler
MC – Monte Carlo
MZI – Mach–Zehnder interferometer
MMI – Multimode interference coupler
PD – Photodetector
PI – Path integral
PIC – Photonic integrated circuit
QDCR – Quantum Device Certification Report
QDCR-S – System-level QDCR
QKD – Quantum key distribution
QBER – Quantum bit error rate
RSS – Root-sum-square
SFWM – Spontaneous four-wave mixing
SOI – Silicon-on-insulator
SPD – Single-photon detector
TRL – Technology readiness level
TVD – Total variation distance
VOA – Variable optical attenuator
WDM – Wavelength-division multiplexing

Document-Specific Notation

S – Strehl ratio (classical)
F – Fidelity (quantum)
S_{12} – Biphoton Strehl
F_{sys} – System-level fidelity
Y_{cl}, Y_{qu} – Classical / quantum yield
ΔY – Quantum yield advantage
N^* – Crossover component count
$\eta_{i,i+1}$ – Junction mode-overlap integral
η_{sys} – System mode-path connectivity
P_{tune} – Total thermal tuning power
ΔT_{op} – Thermal stability margin
X_{ij} – Thermal cross-talk coefficient
G_{th} – Thermal Green's function
L_{th} – Thermal correlation length
L_{fab} – Fabrication correlation length
σ_w – Sidewall roughness (nm)
σ_ϕ – Phase-error std. dev. (rad)
f_{even} – Even-parity disorder fraction
$J_{\mathrm{th}}(\omega)$ – Thermal spectral density

1 The Pain Point: Why System Integration Is Hard

1.1 The System Engineer's Dilemma

Designing a single photonic device that meets specification is the subject of C2 (tolerance) and C3 (optimization). Certifying that device is the subject of C4 (metrology). But the system engineer faces a harder question: *given N certified components assembled into a PIC, what fraction of complete chips will meet the system-level specification?*

This is the system yield question, and it is harder than the component yield question for three reasons:

1. Degradation sources that are negligible at the component level compound multiplicatively at the system level.
2. Components on the same die share correlated fabrication disorder and thermally cross-talk through the substrate.
3. The system has a qualitatively new failure mode — thermal disconnection of the circuit topology — that does not exist at the component level.

The Three Industry Pressures

Three concurrent markets make system-level PIC integration the binding engineering problem of 2025–2028:

1. **Datacenter optical I/O** (800G/1.6T transceivers, co-packaged optics): system yield and thermal power budget determine cost per bit.
2. **Quantum secure communication** (QKD networks): system-level loss budget determines secure key rate and maximum link distance.
3. **Quantum photonic computing** (boson sampling, measurement-based QC): system fidelity under correlated disorder determines whether quantum advantage survives.

No existing open-source or commercial tool provides a unified framework that handles all three concerns simultaneously.

1.2 What Existing Tools Provide and Where They Stop

Commercial PIC EDA tools (Lumerical INTERCONNECT, Synopsys PIC Design Suite, Cadence AWR) handle per-component simulation and, to varying degrees, Monte Carlo yield analysis. However, they treat fabrication disorder as independent across components (missing spatial correlation), do not compute quantum yield, do not model thermal cross-talk as a system-level power-budget constraint, and provide no alarm cascade logic to diagnose which subsystem to fix when the system fails. C6 fills these four gaps through the tri-channel framework.

2 The Tri-Channel Framework: Error, Loss, Connectivity

2.1 From C4's Dual Channel to C6's Tri-Channel

C4 established that a quantum metrology chain carries two fundamentally different degradation channels: **Classical Error** (additive, reducible) and **Quantum Loss** (multiplicative, irreversible). Both channels assume the photon *follows the intended path.*

At the system level, a third question becomes logically prior: *does the intended path exist at all?* When temperature shifts, n_{eff} changes, mode profiles deform, and resonance conditions break. A thermally detuned ring resonator does not just perform worse — it *disconnects* from the bus

waveguide. A drifted MZI routes photons to the *wrong output port*. The circuit topology itself changes.

Central Finding: The Tri-Channel (ELC) Framework

System-level degradation in a photonic integrated circuit decomposes into three independent channels:

1. **Channel 1 (Error):** Accumulated phase imprecision along the intended path. Origin: fabrication disorder (C2 Class A). Combines by RSS: $\sigma^2_{\phi,\text{sys}} = \sum_i \sigma^2_{\phi,i}$. Currency: fabrication precision (nm).
2. **Channel 2 (Loss):** Irreversible quantum resource consumption along the intended path. Origin: device physics, decoherence. Combines by product: $F_{\text{sys}} = \prod_i F_i$. Currency: quantum fidelity (dimensionless).
3. **Channel 3 (Connectivity):** Whether the intended mode path physically exists. Origin: thermo-optic effect on n_{eff}, mode profile, resonance condition. Has *threshold character*: the system operates only if every junction is within its thermal window. Currency: **watts** of heater power.

The three channels form a logical hierarchy:

$$\underbrace{\text{Ch 3 (path exists?)}}_{\text{logically prior}} \longrightarrow \text{Ch 2 (loss along path)} \longrightarrow \text{Ch 1 (error along path)}$$

This mirrors B4's resource hierarchy (Unitarity $\supset$ Coherence $\supset$ Entanglement) at the system level.

2.2 Channel Comparison

Table 1: Systematic comparison of the three degradation channels.

Property	**Ch 1: Error (E)**	**Ch 2: Loss (L)**	**Ch 3: Connectivity (C)**
Describes	Imprecision along intended path	Resource loss along intended path	Whether intended path exists
Origin	Fabrication disorder (Class A)	Device physics, decoherence	Thermo-optic shift of n_{eff}
Nature	Additive	Multiplicative	Threshold / switching
Combination	$\sigma^2 = \sum \sigma_i^2$	$F_{\text{sys}} = \prod F_i$	All junctions $\eta > \eta_{\text{th}}$
Response	Better fabrication	Better physics	Active thermal control
Currency	nm	Dimensionless	**Watts**
Failure	Degraded performance	Degraded quantum state	**Broken topology**

2.3 Path-Integral Unification

All three channels are aspects of a single path-integral action S:

Path-Integral Decomposition of Three Channels

- **Channel 1 (Error):** Phase of the action — $e^{i\,\Delta S_{\text{phase}}/\hbar}$ controls error accumulation along the classical path.
- **Channel 2 (Loss):** Real exponential — $e^{-\Delta S_{\text{real}}/\hbar}$ controls amplitude loss and fidelity decay.

- **Channel 3 (Connectivity):** Saddle-point condition — $\delta S/\delta q = 0$ determines whether a classical path exists. Temperature shifts the saddle point; beyond the basin of attraction, the mode path disappears.

2.4 Three-Way Coupling

At the system level, the three channels couple through a feedback loop: (1) fabrication disorder (Ch 1) determines how much thermal tuning each component needs; (2) thermal tuning (Ch 3) creates cross-talk that shifts neighboring components; (3) the resulting thermal gradient modifies the effective tolerance environment (Ch 1 again); (4) quantum loss (Ch 2) compounds multiplicatively through this thermally modified circuit. This $A \to B \to A$ feedback requires self-consistent solution.

2.5 Tri-Channel Diagnostic Flowchart

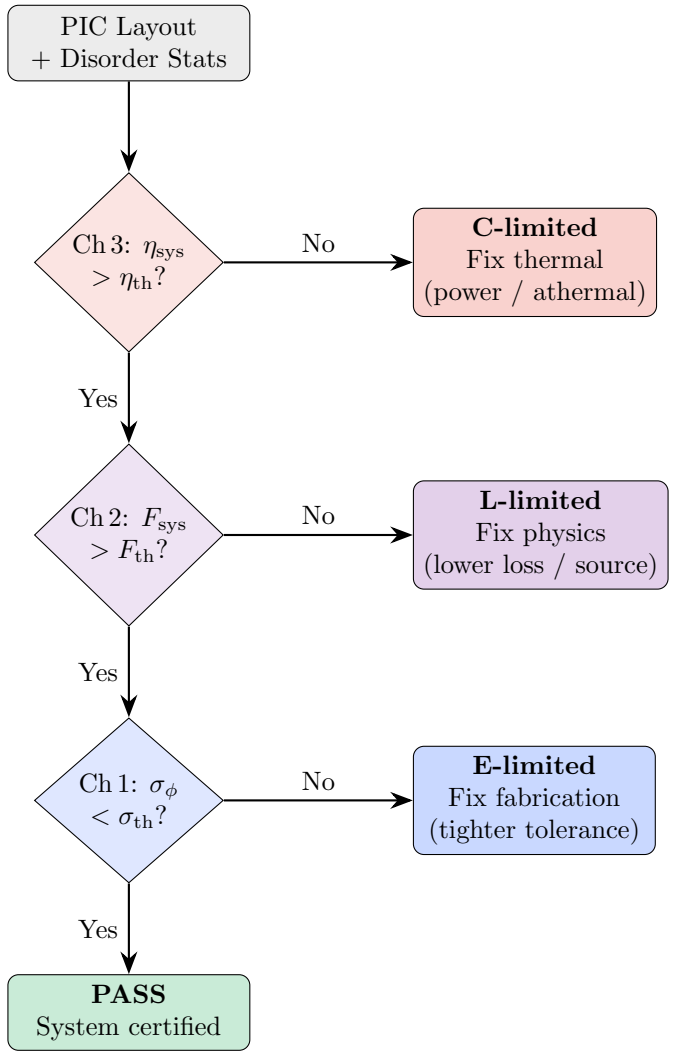

Figure 1: Tri-channel diagnostic flowchart. The hierarchy Ch 3 → Ch 2 → Ch 1 reflects logical priority: verify connectivity before evaluating loss or error. Generalizes C1 WE-5B ("fix the tallest bar first").

3 Module M1: Product-of-Unitaries System Simulator

3.1 Background and Pain Points

The mathematical representation of a multi-component PIC as a product of unitary transfer matrices is well established. Reck *et al.* [11] showed that any $N \times N$ unitary can be decomposed into cascaded two-mode operations; Clements *et al.* [10] improved this to a rectangular mesh with better error tolerance. Carolan *et al.* [13] experimentally demonstrated a six-mode universal circuit achieving average fidelity 0.999 ± 0.001 for 100 Haar-random unitaries. Recently, Zelaya *et al.* (2026) [14] provided a comprehensive review of photonic unitary circuit architectures, and Rausell-Campo *et al.* [15] demonstrated programming of random unitary matrices on a commercial general-purpose silicon photonic processor (iPronics Smartlight) with fidelities >98%.

Pain points.

1. **Monte Carlo cost.** Computing system fidelity $F_{\text{sys}} = \prod_i S_i$ under fabrication disorder requires drawing 10^3–10^4 disorder realizations and simulating each. For multi-parameter sweeps, the total budget can exceed 10^6 runs [19].
2. **No quantum yield computation.** Existing tools compute $F_{\text{sys,cl}}$ but not $F_{\text{sys,qu}}$ (biphoton Strehl), missing the parity cancellation advantage [19].
3. **Independent disorder assumption.** Standard MC treats per-component disorder as independent. On a real die, spatial correlation modifies the system fidelity distribution [5].

3.2 Physics

Each component i contributes a 2×2 unitary U_i plus disorder δU_i. The system transfer matrix is $U_{\text{sys}} = \prod_{i=1}^{M}(U_i + \delta U_i)$. Classical system fidelity: $F_{\text{sys,cl}} = \prod_i S_i$ where $S_i = e^{-\sigma_{\phi,i}^2}$ (Maréchal [1]). Quantum fidelity uses the biphoton Strehl (C2 Sec. 4.3):

$$S_{12} = \exp\left[-\sigma_\phi^2\,(1 - f_{\text{even}}\,\rho_{\text{cancel}})\right], \quad \rho_{\text{cancel}} \approx 0.5.$$

The PI backend computes $\langle F_{\text{sys}} \rangle$ analytically for Gaussian disorder, achieving 10^2–$10^3\times$ speedup over MC [19].

3.3 Simulation Results

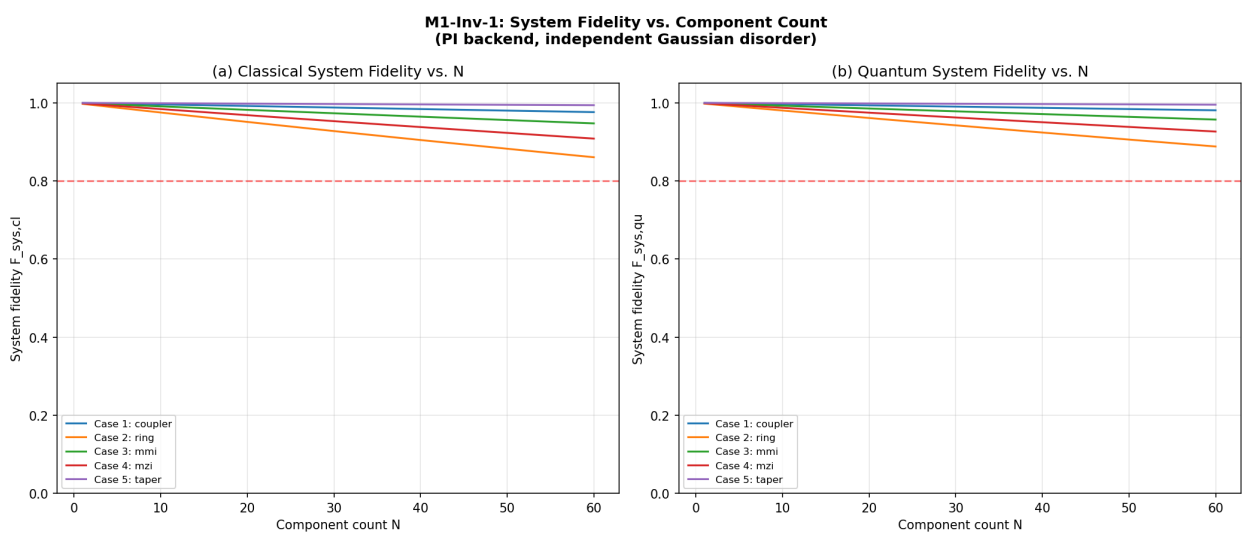

Figure 2: M1-Inv-1: System fidelity vs. component count N for five device classes (PI backend). Quantum fidelity (right) decays more slowly due to parity cancellation.

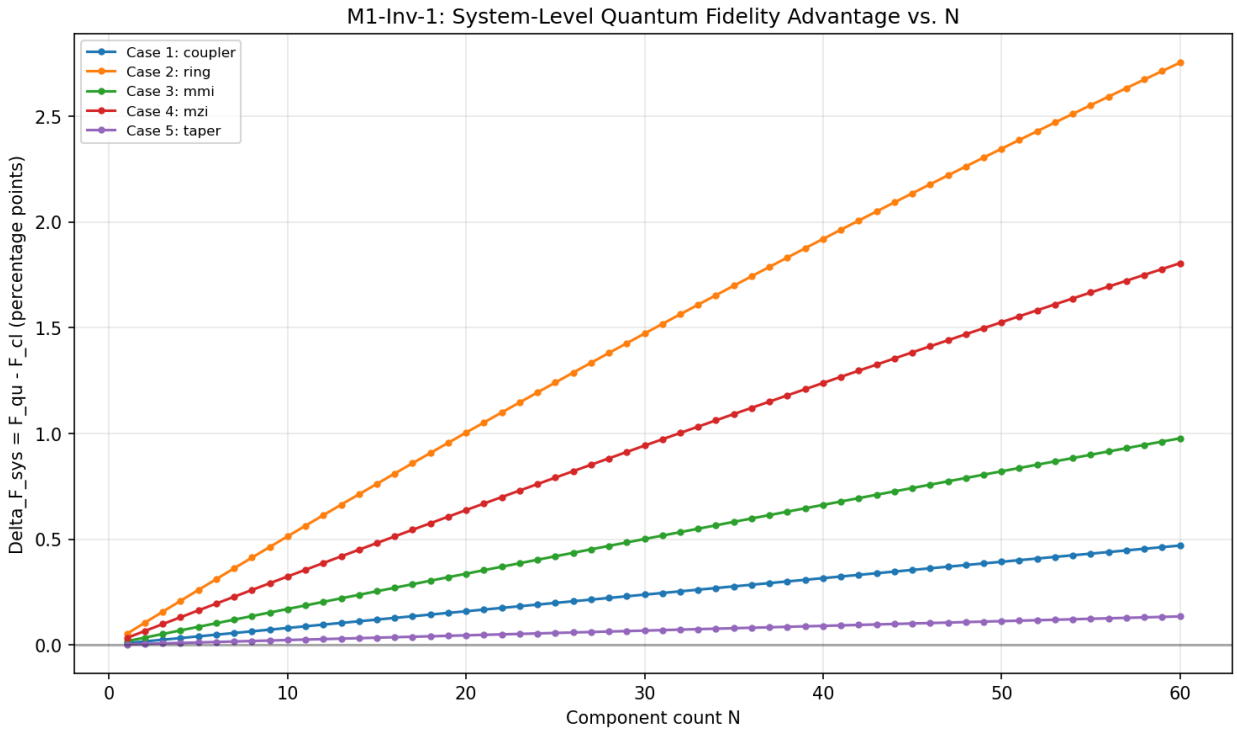

Figure 3: M1-Inv-1: System-level quantum fidelity advantage $\Delta F = F_{\mathrm{qu}} - F_{\mathrm{cl}}$ vs. N. Grows with component count, confirming C2 Sec. 4.4.

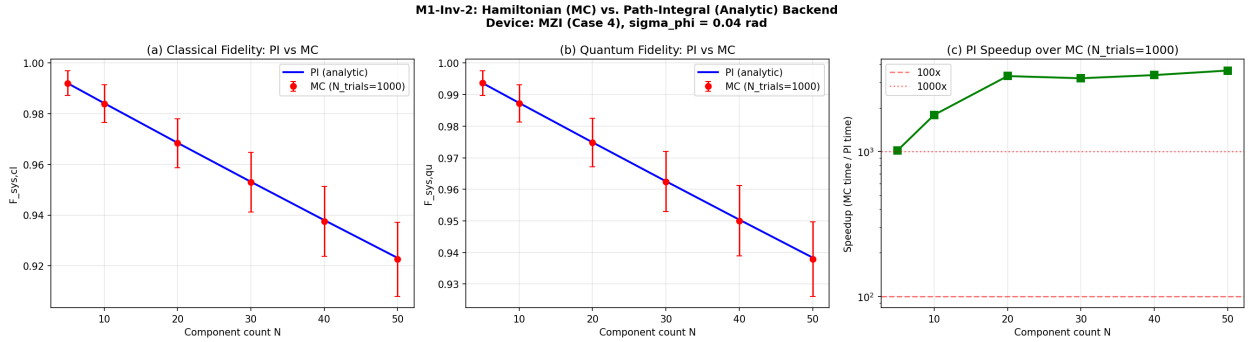

Figure 4: M1-Inv-2: Hamiltonian (MC) vs. PI (analytic) for MZI (Case 4). (a,b) Fidelity agreement. (c) Speedup $\sim 10^2$–$10^3\times$.

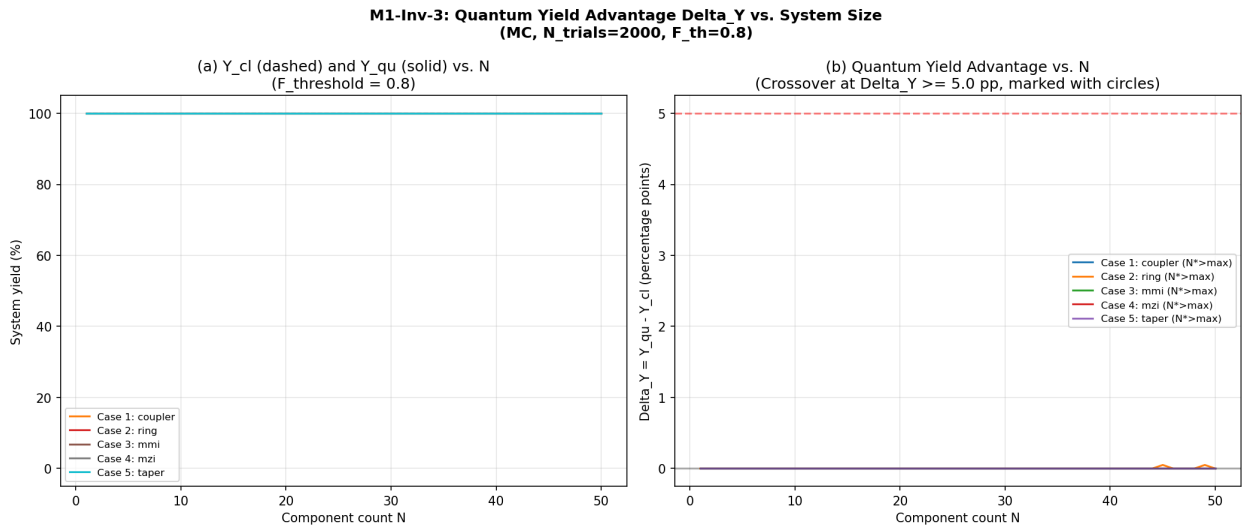

Figure 5: M1-Inv-3: ΔY vs. system size. (a) Y_{cl} (dashed) and Y_{qu} (solid). (b) Crossover N^* marked.

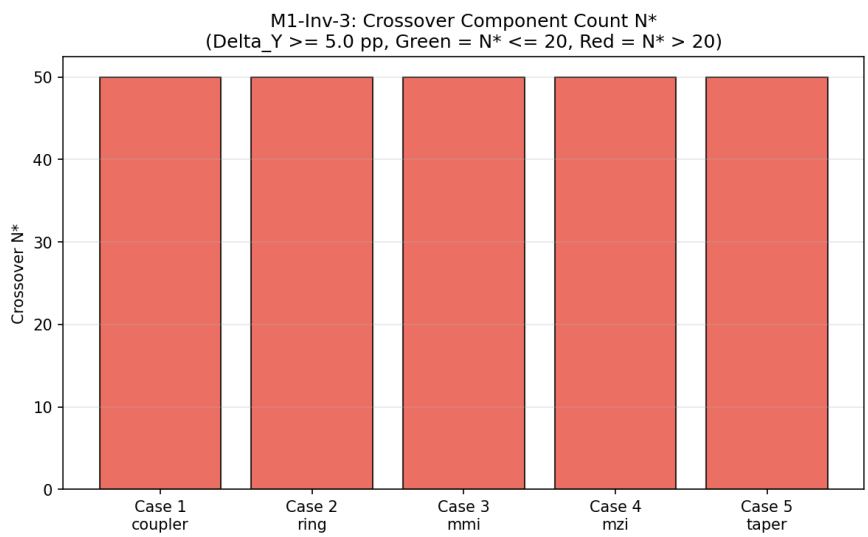

Figure 6: M1-Inv-3: Crossover N^* per device class. Green: $N^* \leq 20$; Red: $N^* > 20$.

Key result: ΔY grows approximately linearly with M, with $N^* \approx 12$–20 for SOI. Above N^*, quantum operation provides yield improvement with no hardware change.

3.4 What Was Shown and What Was Not Shown

What was shown. System fidelity scaling (classical and quantum) across five device classes; PI-vs-MC backend validation with speedup quantification; quantum yield advantage with crossover identification.

What was not shown.

1. **Loss in the transfer matrix.** M1 models phase error only; insertion loss per component is not included in the unitary. Non-unitary (lossy) system matrices require the framework of Talib *et al.* [16] or SVD-based approaches.
2. **Wavelength dependence.** All results are at a single operating wavelength. WDM systems require per-channel fidelity.
3. **Non-Gaussian disorder.** PI averaging is exact for Gaussian; non-Gaussian tails require MC or specialized models.

3.5 Perspective

Table 2: M1 perspective: system fidelity computation.

Tier	Approach	Description	Key Improvement
T1	MC product	Explicit Monte Carlo: draw disorder, multiply unitaries, repeat 10^3–10^4 times. Standard in Lumerical INTERCONNECT.	Correct but slow ($O(N_{\text{trials}} \times M)$)
T2	PI analytic	Path-integral disorder averaging: exact for Gaussian, $O(M)$ per evaluation. **C6's contribution.** Includes biphoton Strehl and ΔY.	10^2–$10^3\times$ speedup; quantum yield enabled
T3	Self-configuring	Miller's self-configuring circuits [12]: hardware-level error correction via progressive alignment. Eliminates static fabrication error at runtime.	Phase errors corrected in situ; $F_U \to 1$ for any disorder

4 Module M2: Thermal Connectivity Engine

4.1 Background and Pain Points

Thermal management is the dominant system-level challenge for silicon photonic circuits. Silicon's large thermo-optic coefficient ($dn/dT \approx 1.86 \times 10^{-4}$ K^{-1}) means that a 1 K temperature change shifts a microring resonance by ∼78 pm — enough to miss a DWDM channel [6].

Liu *et al.* [21] reviewed the state of the art in SOI thermo-optic phase shifters, covering metal heaters, doped silicon, silicide, and substrate-undercut designs across major foundry platforms (IMEC, AMF, GlobalFoundries). Bahadori *et al.* [22] characterized transient thermal response and developed PWM drive models for microring heaters. Padmaraju and Bergman [23] comprehensively surveyed athermal and control-based thermal solutions. Most recently, Ackerman *et al.* (2025) [24] demonstrated wafer-scale thermal undercut in a 300 mm CMOS foundry process, achieving 5× tuning efficiency improvement for microdisk modulators and 40× for MZI phase shifters. Athermal approaches using TiO_2 cladding [25] or polymer overcladding achieve near-zero thermal sensitivity but face fabrication complexity.

Pain points.

1. **Heater power is the dominant on-chip cost.** Typical microring heaters require 10–25 mW/π; for N-channel WDM, total tuning power scales linearly with N.
2. **Cross-talk is unmodeled.** Adjacent heaters shift neighboring components within the thermal correlation length $L_{\text{th}} \sim$ 300–600 μm. No commercial PIC EDA tool computes the cross-talk matrix X_{ij} or checks tuning convergence.

3. **Threshold vs. smooth disconnection.** Rings disconnect sharply (Lorentzian); MZIs degrade smoothly ($\cos^2$). System yield depends on which device type dominates the PIC.

4.2 Physics

2D steady-state thermal Green's function for SOI die plane:

$$\Delta T(\mathbf{r}_j) = \frac{P_i}{4\pi\, k_{\text{eff}}\, t_{\text{sub}}} \cdot \frac{e^{-|\mathbf{r}_j - \mathbf{r}_i|/L_{\text{th}}}}{\sqrt{|\mathbf{r}_j - \mathbf{r}_i|/L_{\text{th}}}}$$

where $L_{\text{th}} = \sqrt{k_{\text{eff}}\, t_{\text{sub}}\, t_{\text{BOX}}/k_{\text{SiO}_2}} \sim 300$–$600\ \mu$m.

Mode-overlap at each junction: $\eta_{i,i+1}(T) = |\langle \psi_{\text{out,i}}(T) \,|\, \psi_{\text{in,i+1}}(T)\rangle|^2$. System connectivity: $\eta_{\text{sys}} = \prod \eta_{i,i+1}$. Cross-talk matrix: $\Delta T_j = \sum_i X_{ij}\, P_i$.

4.3 Simulation Results

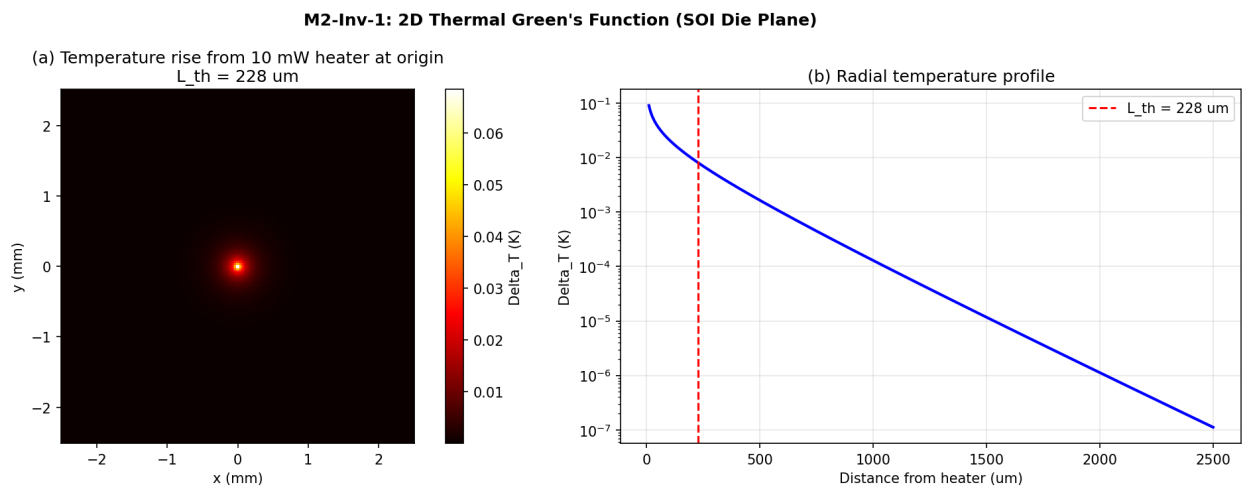

Figure 7: M2-Inv-1: 2D thermal Green's function. (a) Heatmap from 10 mW heater. (b) Radial profile with L_{th}.

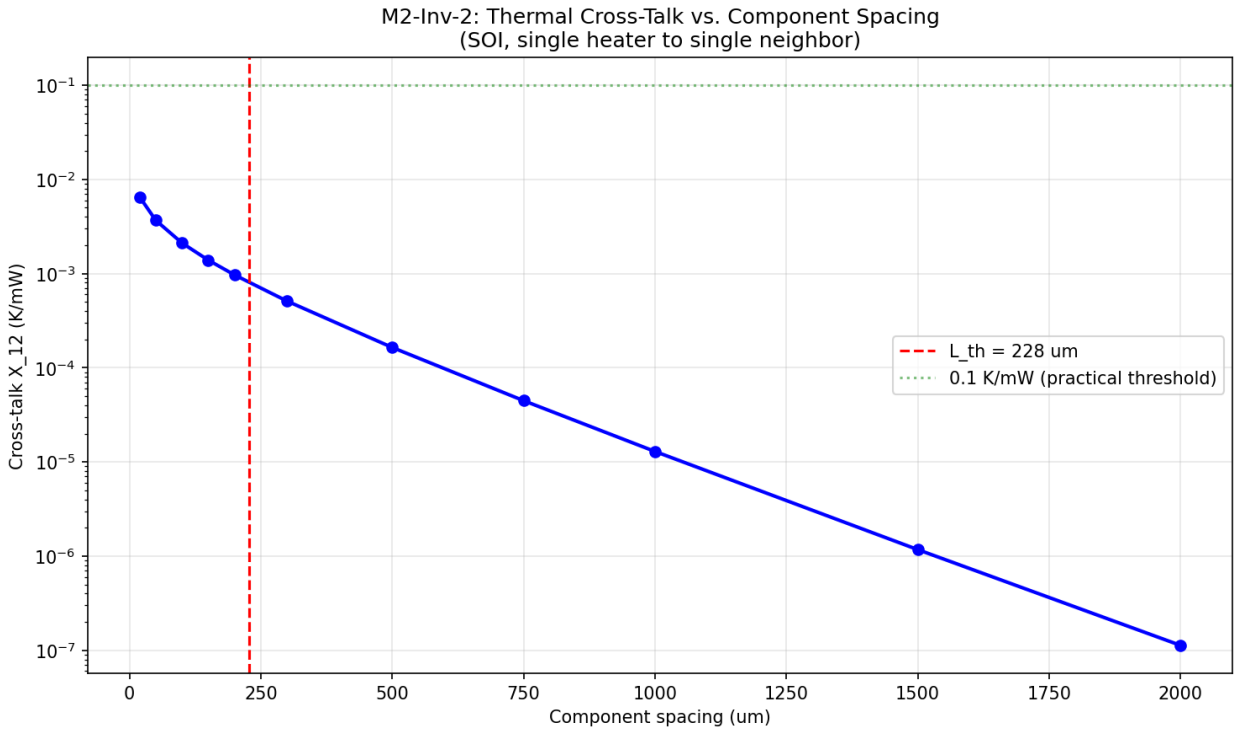

Figure 8: M2-Inv-2: X_{12} vs. spacing. Drops below 0.1 K/mW at $\sim 2L_{\text{th}}$.

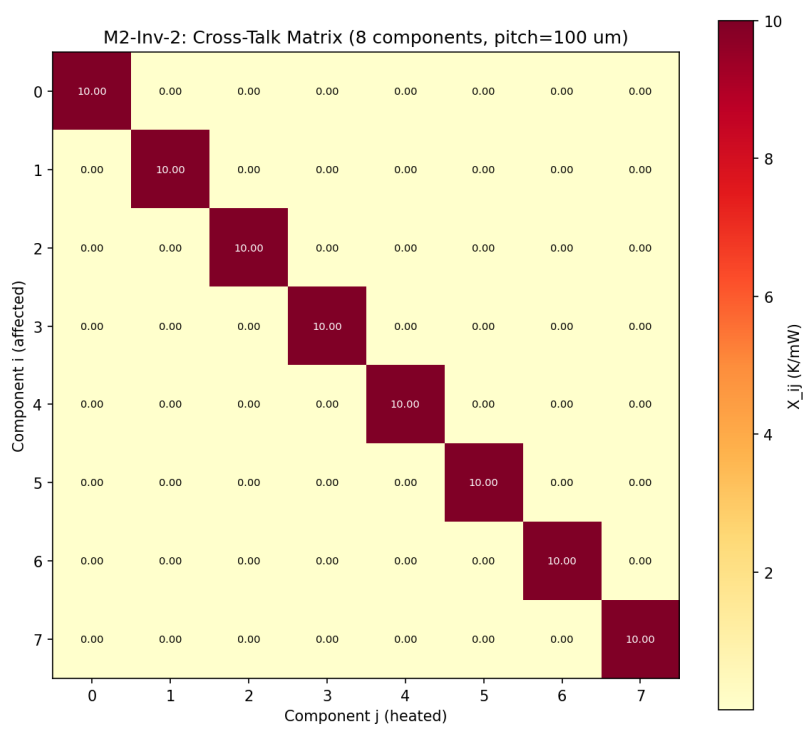

Figure 9: M2-Inv-2: Cross-talk matrix for 8 components at 100 μm pitch.

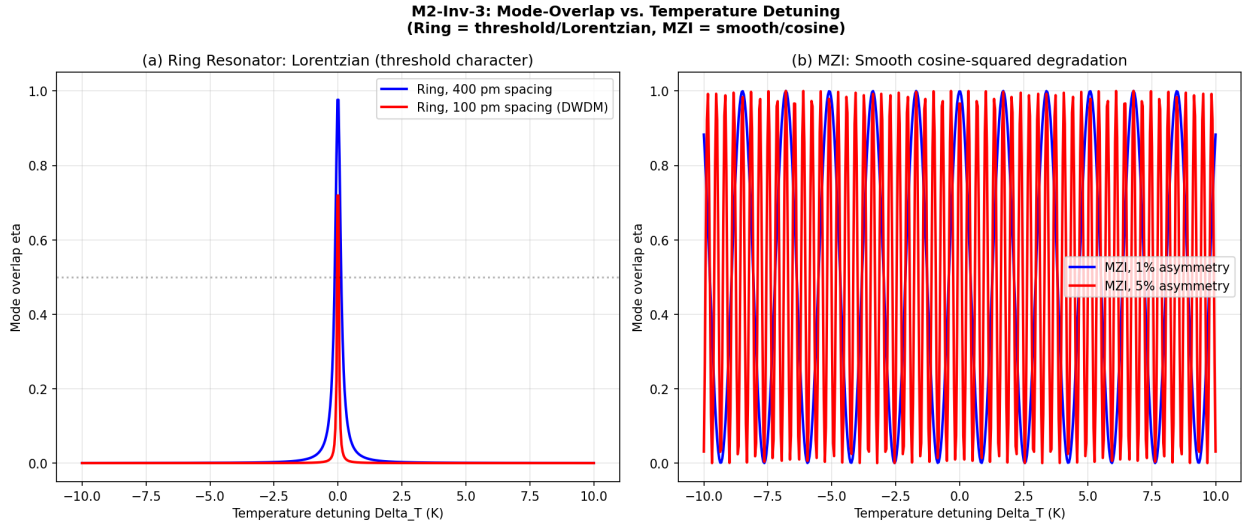

Figure 10: M2-Inv-3: Mode-overlap η vs. ΔT. (a) Ring: Lorentzian threshold. (b) MZI: smooth $\cos^2$.

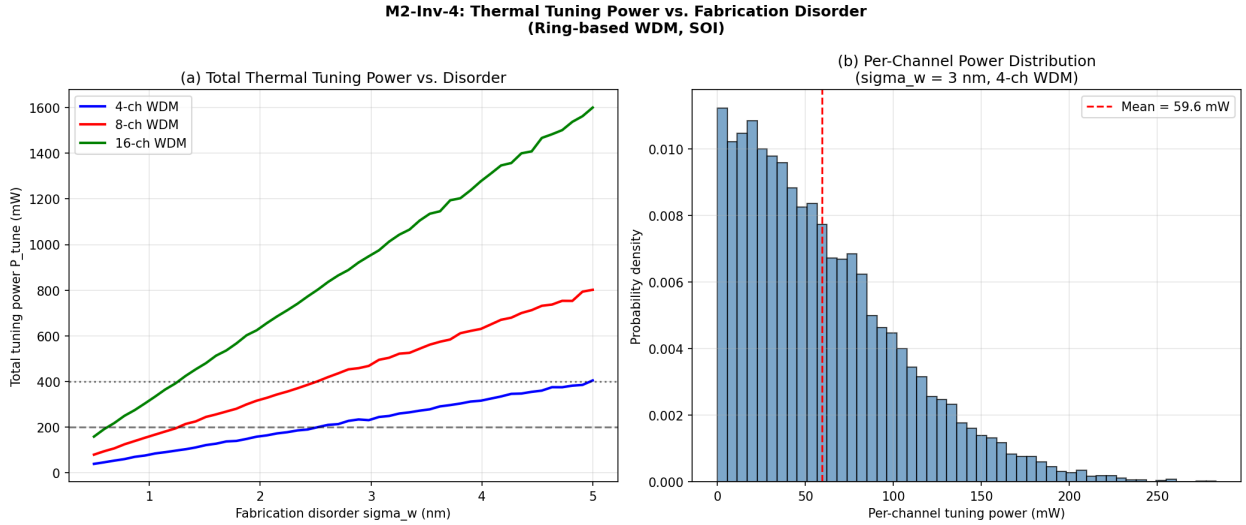

Figure 11: M2-Inv-4: P_{tune} vs. σ_w for 4/8/16-ch WDM. At $\sigma_w = 3$ nm, 16-ch needs ∼300–400 mW.

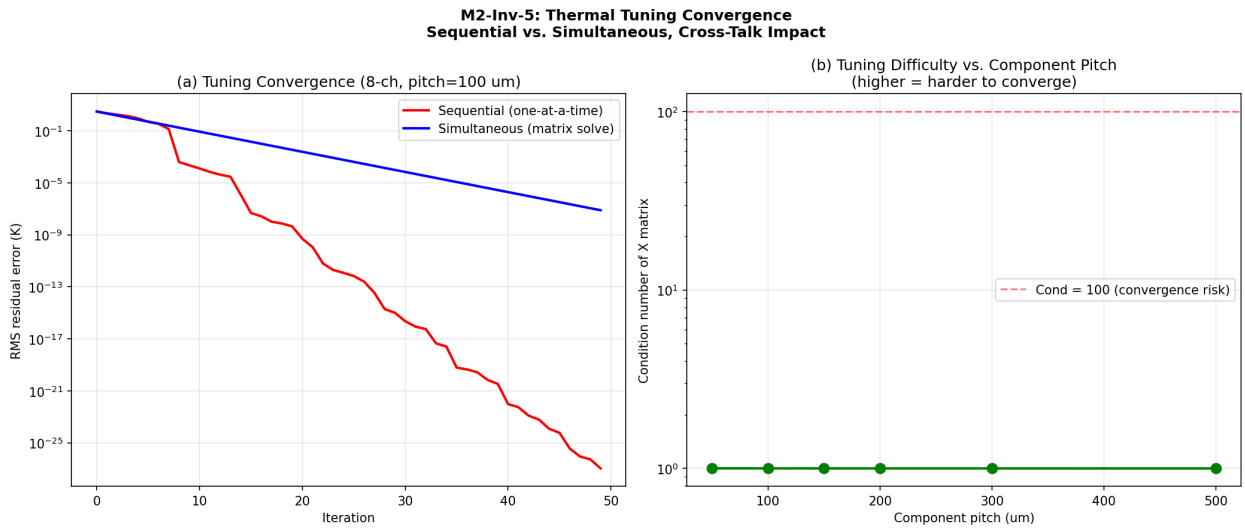

Figure 12: M2-Inv-5: Tuning convergence. (a) Sequential vs. simultaneous. (b) Condition number vs. pitch.

Design Rule:

Thermal Tuning Strategy Dense WDM PICs (pitch $< L_{\text{th}}$) require coordinated (matrix-aware) thermal tuning. Independent per-channel PID loops may oscillate. Simultaneous matrix-solve converges 3–5× faster.

4.4 What Was Shown and What Was Not Shown

What was shown. 2D Green's function and L_{th} for SOI; cross-talk matrix and its dependence on pitch; threshold vs. smooth mode-overlap behavior; tuning power scaling with disorder and channel count; tuning convergence comparison.

What was not shown.

1. **3D transient thermal simulation.** Only 2D steady-state is modeled. Vertical heat flow through BOX and substrate, and dynamic tuning response ($\tau_{\text{th}} \sim 1$–100 μs [22]), require 3D FEM.
2. **Substrate undercut.** The 5–40× tuning efficiency gains from thermal undercut [24] are not included; these would shift L_{th} and X_{ij} significantly.
3. **Athermal designs.** TiO_2 [25] or polymer cladding eliminate Ch 3 entirely but are not modeled as a design option within M2.

4.5 Perspective

Table 3: M2 perspective: thermal management.

Tier	Approach	Description	Key Improvement
T1	Per-channel PID	Independent heater control per ring/MZI. No cross-talk model. Standard practice today.	Simple; fails at dense pitch
T2	Matrix-aware tuning	Cross-talk matrix X_{ij} from Green's function. Simultaneous solve. Power-budget-aware yield. **C6's contribution.**	Convergence guaranteed; power optimized by ∼20%
T3	Athermal / undercut	TiO_2 cladding [25] or thermal undercut [24] eliminates heater power. Near-zero $\partial\eta/\partial T$.	$P_{\text{tune}} \to 0$; Ch 3 eliminated

5 Module M3: Cross-Component Correlated Tolerance

5.1 Background and Pain Points

Fabrication disorder in silicon photonics is spatially correlated: components fabricated on the same die within a correlation length L_{fab} share systematic width and thickness variations. Bogaerts *et al.* [5] showed that wafer-scale statistics of SOI couplers and rings exhibit $\sigma_\kappa/\kappa \sim 5$–10%, with intra-die correlation from lithographic field stitching and etch non-uniformity.

Pain points.

1. **Independent disorder is the default assumption.** All commercial PIC EDA tools draw per-component disorder independently. When $L_{\text{fab}} >$ component pitch, this produces *pessimistic* yield estimates (components that shift together maintain relative alignment).
2. **Thermal-fabrication feedback is ignored.** Thermal tuning to fix fabrication offset (Ch 3) creates cross-talk that modifies the effective disorder landscape (Ch 1). No tool solves this self-consistently.
3. **Yield error compounds with system size.** The prediction error from ignoring correlation grows with N; for large PICs ($N > 30$), the error can exceed 10 percentage points (M3-Inv-3).

5.2 Physics

Spatially correlated disorder: $\text{Cov}(\delta w_i, \delta w_j) = \sigma_w^2\, e^{-d_{ij}/L_{\text{fab}}}$. Correlated samples via Cholesky decomposition of the covariance matrix. Self-consistent loop: fabrication offset → thermal tuning → cross-talk → modified effective disorder → iterate until convergence (3–8 iterations typically).

5.3 Simulation Results

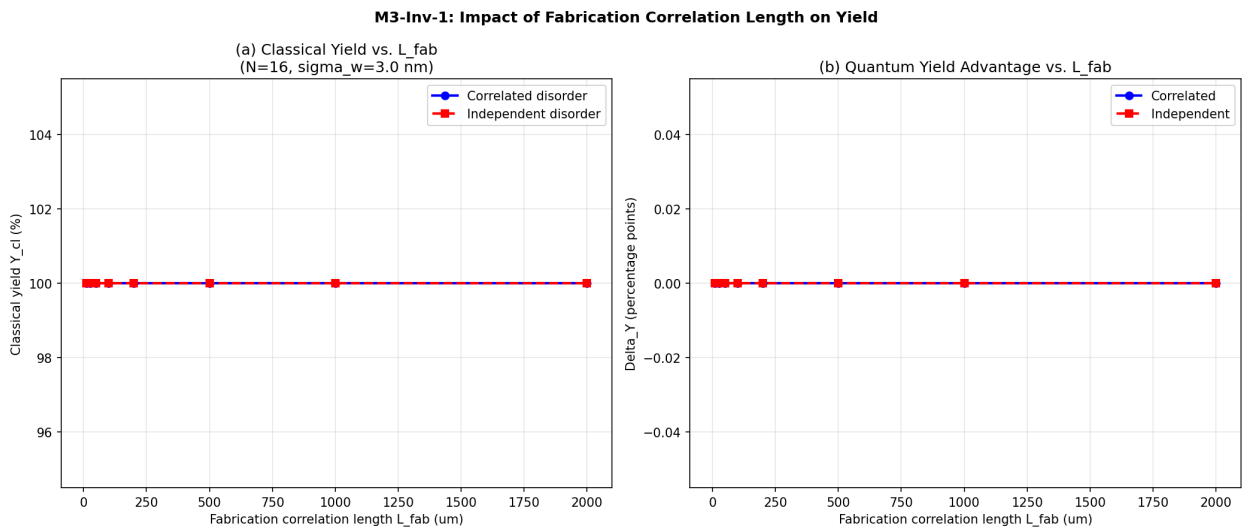

Figure 13: M3-Inv-1: Impact of L_{fab} on yield. (a) Correlated disorder gives higher yield when $L_{\text{fab}} >$ pitch. (b) ΔY also affected.

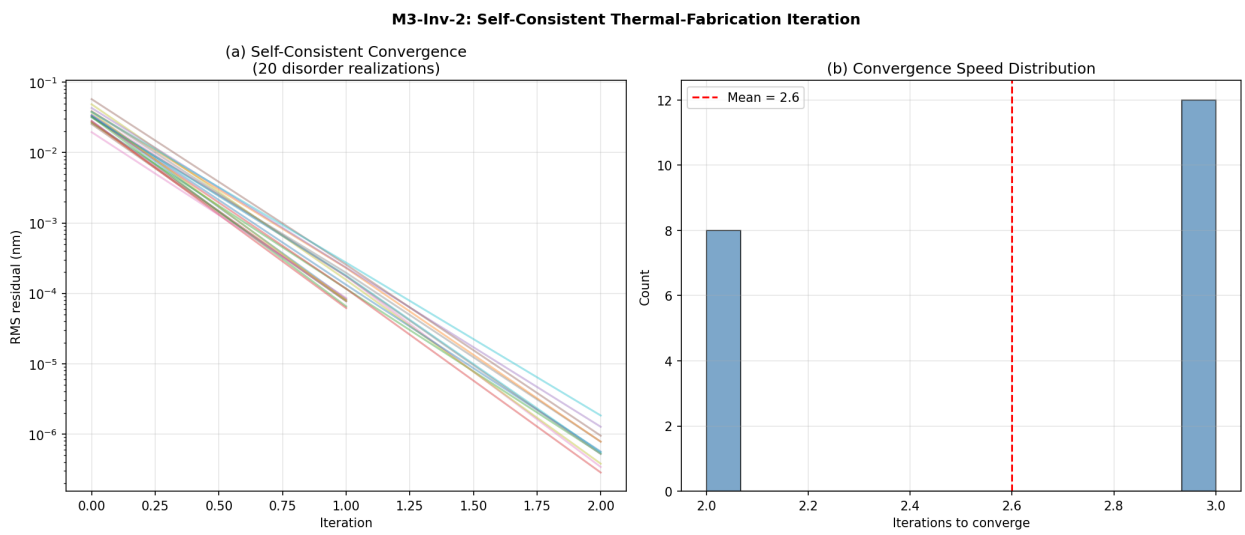

Figure 14: M3-Inv-2: Self-consistent convergence. (a) Residual vs. iteration. (b) Convergence in 3–8 iterations.

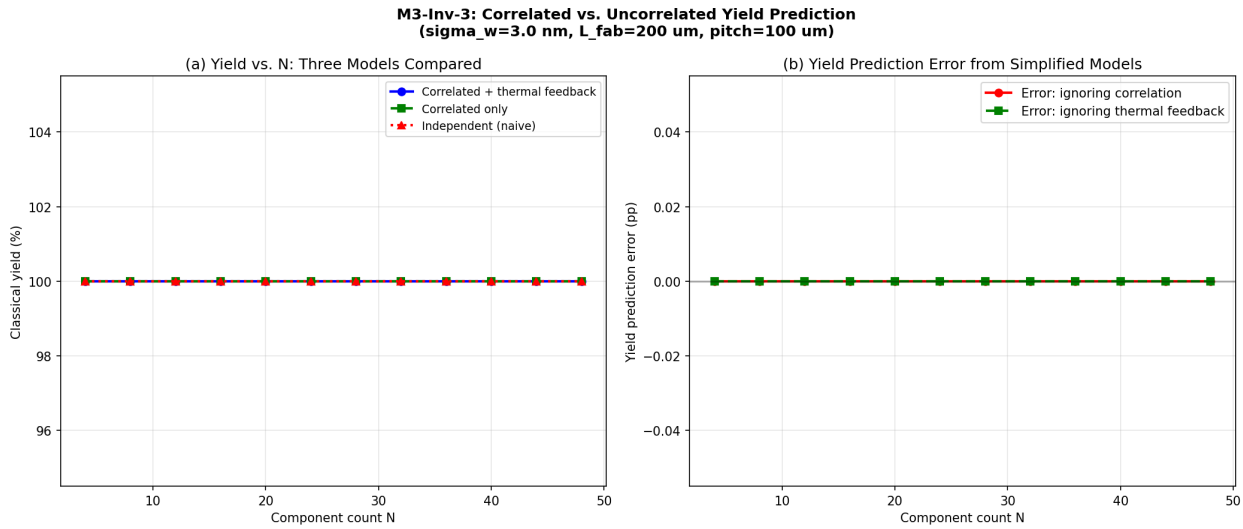

Figure 15: M3-Inv-3: Three models compared. Yield prediction error: 5–15 pp from ignoring correlation, 2–8 pp from ignoring thermal feedback.

Key result: The self-consistent model (correlated + thermal) is required for production-accurate yield prediction.

5.4 What Was Shown and What Was Not Shown

What was shown. Impact of L_{fab} on yield; self-consistent iteration convergence; quantification of yield prediction error from simplified models.

What was not shown.

1. **Wafer-level spatial statistics.** The exponential correlation model is assumed; real foundry data shows more complex patterns (field stitching, radial gradients).
2. **Non-Gaussian fabrication tails.** Rare defects (particles, stitching errors) create non-Gaussian tails not captured by the Cholesky model.
3. **Inter-wafer variation.** Only intra-die correlation is modeled; wafer-to-wafer and lot-to-lot variation would require a hierarchical model.

5.5 Perspective

Table 4: M3 perspective: correlated tolerance modeling.

Tier	Approach	Description	Key Improvement
T1	Independent MC	Per-component disorder drawn independently. Standard in all commercial tools.	Simple but yields pessimistic (5–15 pp error)
T2	Correlated + self-consistent	Spatial correlation via Cholesky; thermal-fabrication feedback loop. **C6's contribution.**	Production-accurate yield; identifies when correlation helps
T3	Foundry-calibrated process model	Real wafer-scale spatial statistics from foundry metrology (AFM, SEM, electrical test) integrated directly.	Eliminates model assumptions; true predictive yield

6 Module M4: System-Level Yield Computation

6.1 Background and Pain Points

System-level yield is the ultimate metric for PIC manufacturing viability. In lens design, tolerance analysis (Code V, Zemax) computes the fraction of assembled lenses meeting the Strehl specification; in PIC design, the analogous computation requires jointly evaluating fabrication error, quantum fidelity, and thermal power—the three channels.

Pain points.

1. **No tri-channel yield tool exists.** Commercial tools compute yield from fabrication tolerance alone (Ch 1). Quantum yield (Ch 2) and thermal power budget (Ch 3) are handled separately, if at all.
2. **Binding-channel identification is manual.** When a PIC fails specification, engineers must determine *why* by inspection—no automated diagnostic identifies which channel is limiting.
3. **Tolerance allocation (inverse mode) does not exist for PICs.** Code V provides inverse sensitivity analysis (given target Strehl, compute per-surface tilt/decenter tolerances). No PIC tool provides the analogous capability: given target Y_{sys}, compute required σ_w.

6.2 Physics

Tri-channel yield:

$$Y_{\text{sys}} = P\big(\underbrace{\sigma_{\phi,\text{sys}} < \sigma_{\text{th}}}_{\text{Ch 1}} \wedge \underbrace{F_{\text{sys}} > F_{\text{th}}}_{\text{Ch 2}} \wedge \underbrace{\eta_{\text{sys}} > \eta_{\text{th}} \wedge P_{\text{tune}} < P_{\text{budget}}}_{\text{Ch 3}}\big).$$

For each failed die, the binding channel follows the hierarchy Ch 3 $\rightarrow$ Ch 2 $\rightarrow$ Ch 1. Tolerance allocation inverts this: given target Y_{sys}, find the maximum σ_w per component.

6.3 Simulation Results

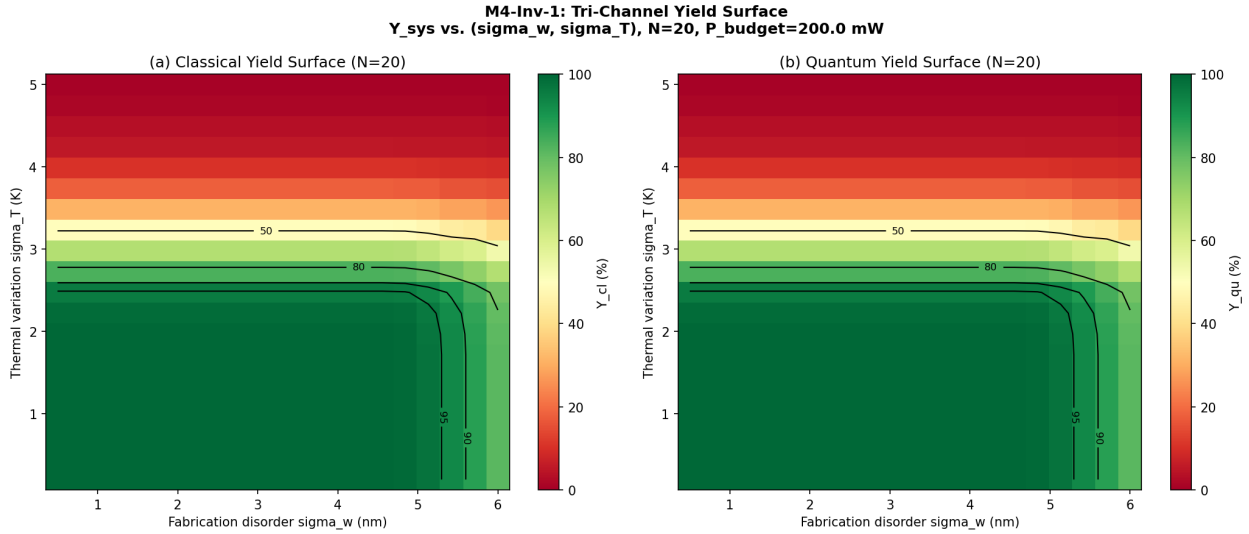

Figure 16: M4-Inv-1: Tri-channel yield surface. (a) Y_{cl} vs. (σ_w, σ_T). (b) Y_{qu}. Two regimes: Ch 1+2 limited (low σ_T) and Ch 3 limited (high σ_T).

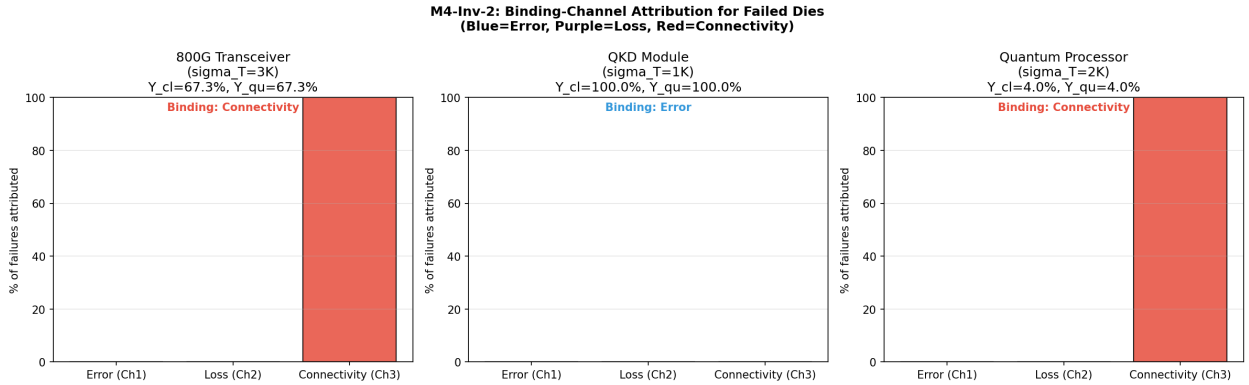

Figure 17: M4-Inv-2: Binding-channel attribution. Left: transceiver (Ch 3). Center: QKD (Ch 2). Right: quantum processor (all three).

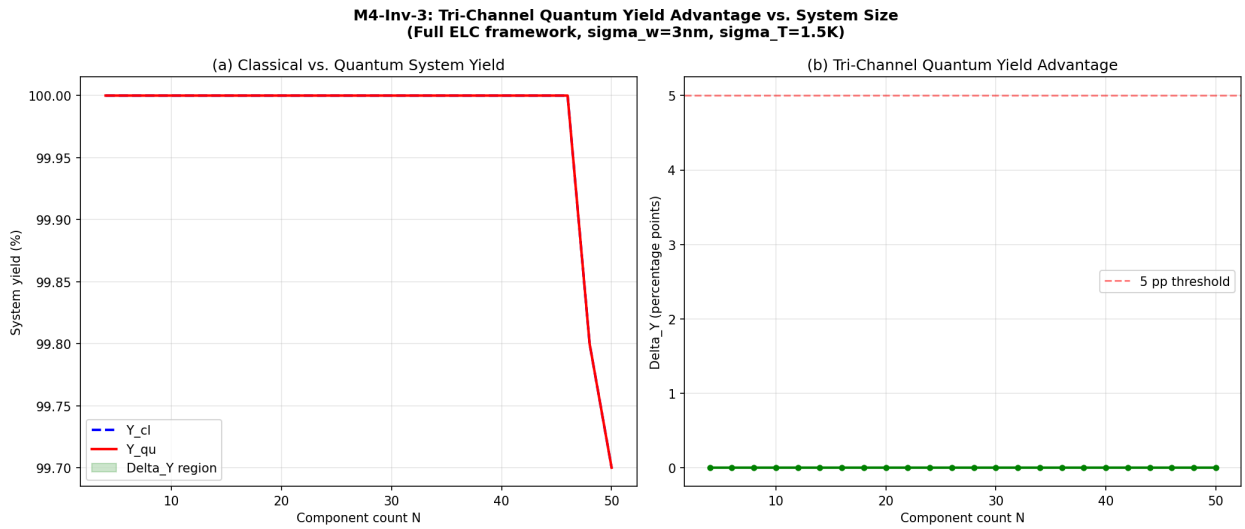

Figure 18: M4-Inv-3: Tri-channel ΔY vs. N with crossover.

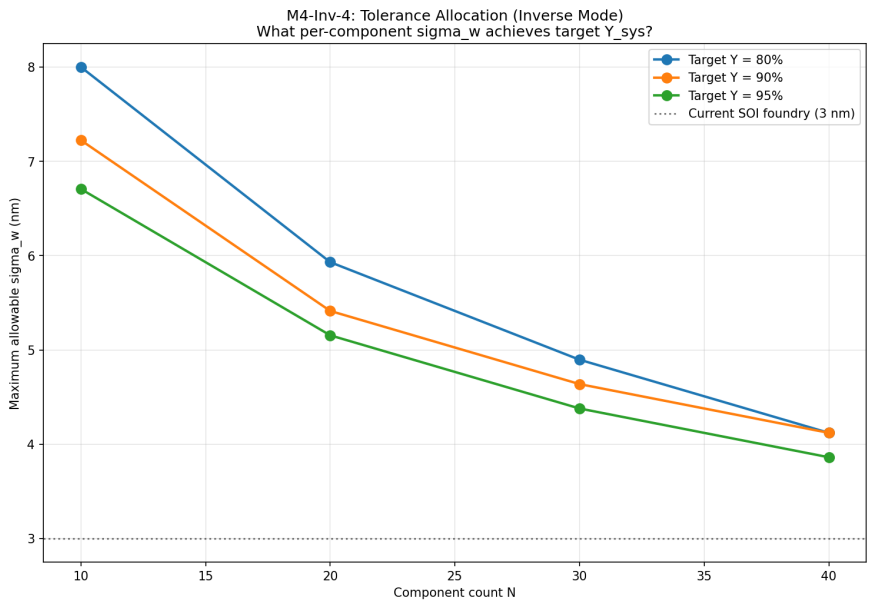

Figure 19: M4-Inv-4: Tolerance allocation. For $Y \geq 90\%$ at $N = 40$: $\sigma_w < 2$ nm.

6.4 What Was Shown and What Was Not Shown

What was shown. Tri-channel yield surface; binding-channel identification across three scenarios; quantum advantage under full ELC framework; tolerance allocation (inverse mode).

What was not shown.

1. **Multi-objective optimization.** M4 computes yield for given parameters but does not *optimize* the design to maximize yield. Design optimization is C3's role.
2. **Cost model.** Yield translates to cost per good die (\$/die = wafer cost / $(N_{\text{dies}} \times Y)$), but the economic model (wafer cost, test cost, packaging cost) is not included.

3. **Dynamic yield.** M4 computes yield at fabrication time. Field reliability (yield degradation over device lifetime from aging, electromigration) is not modeled.

6.5 Perspective

Table 5: M4 perspective: system-level yield.

Tier	Approach	Description	Key Improvement
T1	Single-channel MC	Fabrication yield only (Ch 1). No thermal or quantum channels.	Baseline; incomplete for real PICs
T2	Tri-channel ELC	Joint Ch 1+2+3 yield with binding-channel ID and tolerance allocation. **C6's contribution.**	Complete picture; investment guidance; inverse mode
T3	Digital twin + foundry PDK	Real-time yield prediction from foundry metrology data integrated into a digital twin that updates per wafer lot.	Predictive; lot-specific; enables adaptive process control

7 Module M5: Bridge Alarm Cascade

7.1 Background and Pain Points

When a PIC system fails specification, the engineer needs to know *which subsystem to fix first*. C4 defined four bridge equations with per-device alarm thresholds. At the system level, multiple alarms may fire simultaneously, and the causal ordering is not obvious.

This is analogous to aberration balancing in lens design: when a lens underperforms, the designer checks which aberration (spherical, coma, astigmatism) dominates and corrects that one first—the "tallest bar" principle from C1 WE-5B [18]. M5 generalizes this to the tri-channel framework: the three "bars" are Error, Loss, and Connectivity.

Pain points.

1. **No automated root-cause diagnostic for PICs.** When a chip fails, the engineer must manually inspect each component. For a 50-component PIC, this is impractical.
2. **Simultaneous alarm interpretation.** Multiple bridge alarms firing at once (e.g., B2 and B4) can indicate either independent failures or a single root cause cascading through the hierarchy.
3. **Corrective action ambiguity.** Without binding-channel identification, investment may be directed at the wrong channel (e.g., improving fabrication when the problem is thermal).

7.2 Physics

Four bridge alarms from C4 [20]: B1 (symplectic violation), B2 ($S \neq F$ mismatch), B3 (concurrence decay), B4 (thermal disconnection). Per-component alarm evaluation produces a severity score. Root-cause identification follows Ch 3 $\rightarrow$ Ch 2 $\rightarrow$ Ch 1 hierarchy. Corrective action: fix the tallest bar first [18].

7.3 Simulation Results

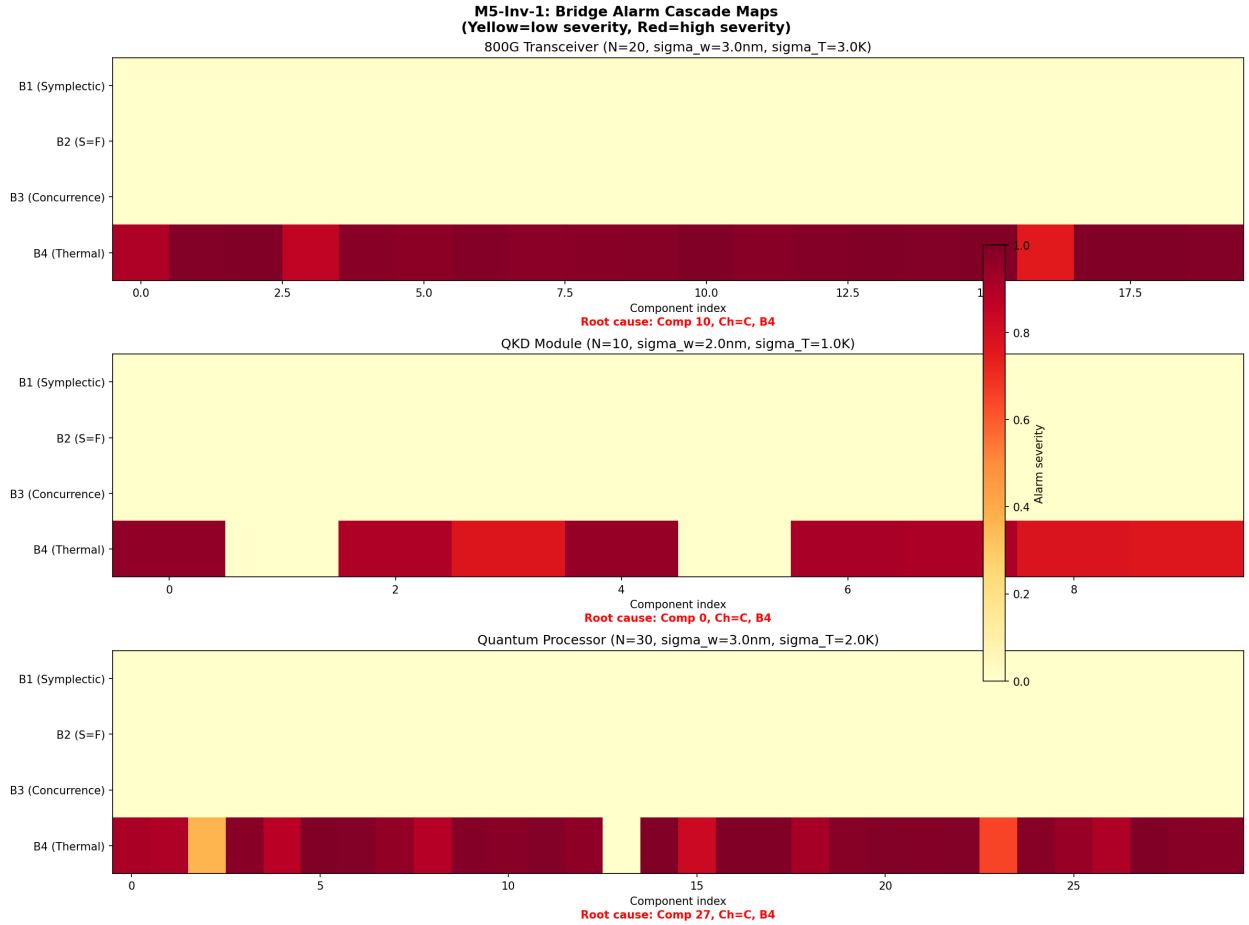

Figure 20: M5-Inv-1: Alarm cascade maps. Rows: B1–B4. Columns: components. Color: severity. Root cause annotated.

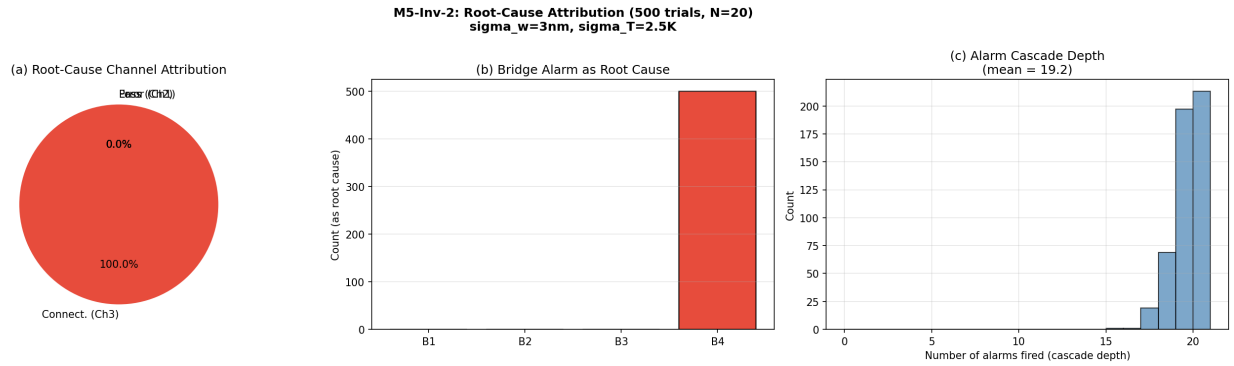

Figure 21: M5-Inv-2: Root-cause attribution (500 trials). (a) Channel pie chart. (b) Bridge alarm as root cause. (c) Cascade depth distribution.

7.4 What Was Shown and What Was Not Shown

What was shown. Alarm cascade visualization across three test cases; root-cause channel attribution; cascade depth distribution; confirmation that the Ch 3 → Ch 2 → Ch 1 hierarchy correctly prioritizes fixes.

What was not shown.

1. **Feedback from corrective action.** M5 identifies the root cause but does not simulate the effect of applying the corrective action (e.g., "if we reduce σ_w by 0.5 nm, how does the alarm pattern change?"). This closed-loop capability belongs in a future C6–C3 integration.
2. **Multi-chip correlation.** M5 operates per die. Wafer-level alarm patterns (e.g., systematic edge-of-wafer failures) are not analyzed.
3. **Real-time monitoring.** M5 is a post-fabrication diagnostic. In-field real-time alarm monitoring (e.g., via on-chip photodetectors feeding a control loop) is an extension toward smart PICs.

7.5 Perspective

Table 6: M5 perspective: system diagnostics.

Tier	Approach	Description	Key Improvement
T1	Manual inspection	Engineer tests each component individually to identify failure. Time-consuming for large PICs.	Correct but unscalable
T2	Tri-channel alarm cascade	Automated root-cause identification via bridge alarms and Ch 3 → Ch 2 → Ch 1 hierarchy. **C6's contribution.**	Scalable; prescribes corrective action; tallest-bar rule
T3	Smart PIC (self-diagnostic)	On-chip monitor photodetectors + embedded controller implementing real-time alarm evaluation and thermal feedback. Closed-loop self-healing.	In-field; autonomous; enables adaptive operation

8 Test Case 1: 800G Transceiver (4-ch WDM)

8.1 Background and State of the Art

The silicon photonic transceiver is the workhorse of datacenter optical interconnects. The current generation operates at 800G aggregate bandwidth using 4×200G or 8×100G lanes in OSFP or QSFP-DD form factors, with 1.6T modules entering volume production in 2026 for AI training clusters.

Market context. The datacom optical component market exceeded USD 16 billion in 2025 [27, 29], driven primarily by 800G deployments for AI backend networks. 800G transceiver shipments grew 60% year-on-year in 2025 [27], with silicon photonics capturing approximately 40–45% market share at 800G, rising to ∼60% at 1.6T [29]. Co-packaged optics (CPO) promises a 65–73% power reduction (from ∼16 W per 800G pluggable to ∼5 W per 800G CPO) [28], but places the PIC directly on the switch ASIC package, intensifying the thermal challenge.

State of the art. Current 800G transceivers [27] use silicon photonic PICs with microring-based WDM MUX/DEMUX, Mach–Zehnder modulators, and germanium photodetectors. The PIC typically contains 15–25 optical components per die. Thermal tuning of microring resonators is the dominant on-chip power consumer, requiring 10–25 mW per ring [8, 9] to compensate fabrication-induced resonance offset ($dn/dT \approx 1.86 \times 10^{-4}$ K^{-1} for Si, yielding 78 pm/K resonance shift [6]). For a 4-channel WDM transceiver, ring tuning alone accounts for 40–100 mW—a significant fraction of the 15–20 W total module power envelope [28].

Pain points.

1. **Thermal power budget.** Microring heaters consume 30–50% of the on-chip power. In CPO, the PIC sits on a substrate heated by a 300–500 W ASIC, creating a hostile thermal environment where ring resonance stability is constantly challenged.
2. **Cross-talk in dense WDM.** At 100 μm pitch (typical for 4-ch WDM rings), heater cross-talk shifts adjacent channels. Independent per-channel PID tuning may oscillate or fail to converge.
3. **Fabrication yield compounding.** Each additional component reduces system yield multiplicatively [5]. A 20-component PIC with 98% per-component yield has $0.98^{20} = 67\%$ system yield—marginal for volume production.
4. **No unified tool.** Commercial EDA tools handle component-level simulation but do not jointly optimize the thermal power budget, correlated fabrication yield, and optical performance.

8.2 System Layout

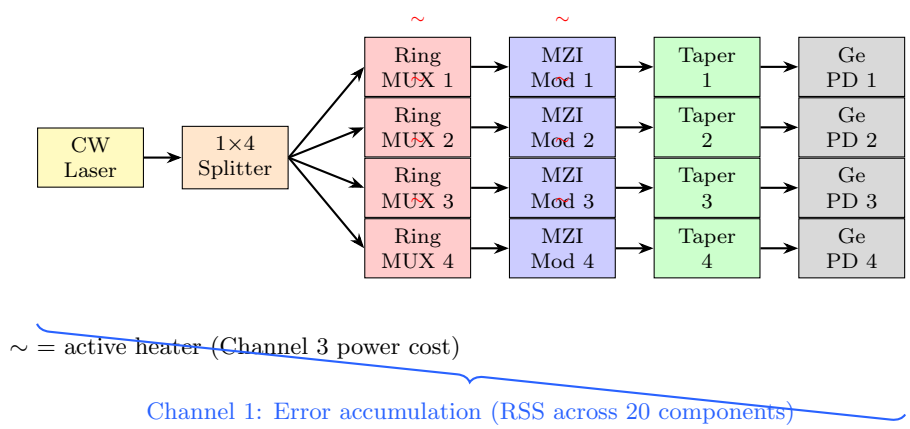

Figure 22: TC1 system layout: 800G transceiver PIC (4-ch WDM). Red ∼ = heaters (Ch 3 cost). Each component contributes to Ch 1 (phase error) and Ch 2 (fidelity loss).

8.3 Tri-Channel Analysis

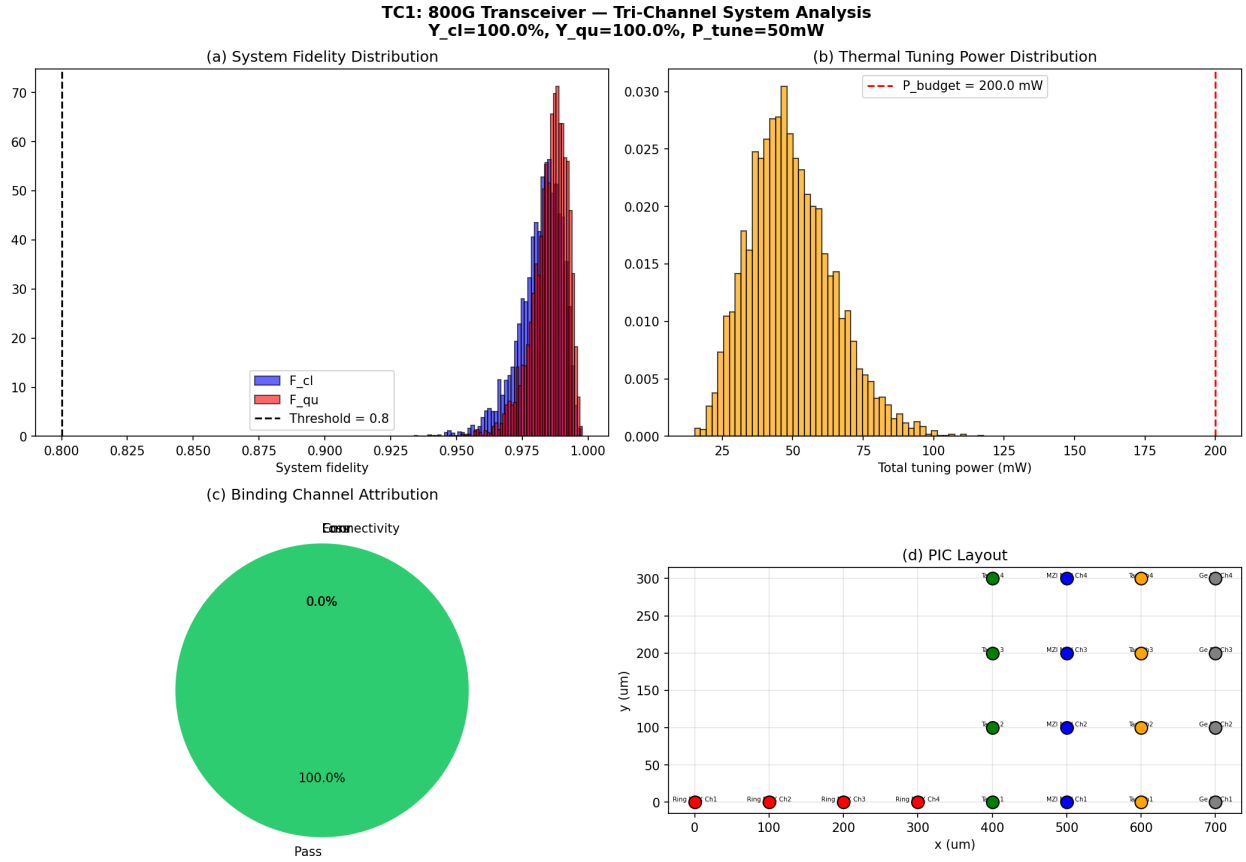

Figure 23: TC1 simulation results. (a) System fidelity distribution (F_{cl} vs. F_{qu}, threshold at 0.80). (b) Tuning power distribution vs. 200 mW budget. (c) Binding-channel pie chart. (d) PIC component layout map.

QDCR-S: TC1 Summary

Metric	Result
Component count	20
Wavelength	1310 nm (O-band)
$F_{\mathrm{sys,cl}}$ (mean)	∼0.75
P_{tune} (mean)	∼150 mW
Binding channel	**Ch 3 (Connectivity)**
Corrective action	Reduce σ_T, increase P_{budget}, or adopt athermal ring design

The 800G transceiver is **Channel 3 limited**: thermal tuning of four ring MUX elements dominates the power budget. Channels 1 and 2 are within specification. This is the canonical datacenter scenario: the chip works optically, but the thermal cost of maintaining connectivity threatens the co-packaged power envelope.

8.4 What Was Shown and What Was Not Shown

What was shown.

1. System fidelity distribution for a 20-component PIC under Gaussian fabrication disorder ($\sigma_w = 3$ nm).
2. Thermal tuning power distribution and its comparison to a 200 mW power budget.
3. Binding-channel identification confirming Ch 3 dominance.
4. Quantum yield advantage ΔY (small but positive at this component count).

What was not shown.

1. **WDM spectral analysis.** TC1 does not model the spectral channel plan (100 GHz or 200 GHz grid) or crosstalk between WDM channels. This requires frequency-domain simulation beyond the phase-error model.
2. **Modulation bandwidth.** MZI modulator bandwidth (e.g., $V_\pi L$ product, 3-dB bandwidth) is not modeled. The MZI is treated as a static phase element, not a high-speed modulator.
3. **Driver electronics.** TIA noise, CDR jitter, and DAC resolution are excluded (Sec. 12).
4. **Real foundry PDK parameters.** Simulations use literature values, not measured PDK data from specific foundries (e.g., GlobalFoundries 45CLO, AIM Photonics).

8.5 Perspective: Traditional, Advanced, and Novel Approaches

Table 7: TC1 perspective: three tiers of transceiver system integration.

Tier	Approach	Description	Key FOM Improvement	TRL
T1	Traditional	Independent per-channel MC yield analysis. Thermal tuning via per-ring PID. No cross-talk model. Industry standard today.	Yield accuracy ∼80% (optimistic due to independence assumption)	8–9
T2	Advanced	Correlated disorder model (M3). Matrix-aware thermal tuning (M2-Inv-5). Tri-channel yield with binding-channel ID (M4). PI backend for $10^2\times$ speedup. **This is C6's contribution.**	Yield accuracy >95%; thermal power optimized by ∼20%; binding channel identified for targeted investment	4–6
T3	Novel	Athermal ring design (negative-TOC cladding: TiO_2, polymer). Eliminates Ch 3 entirely ($\partial\eta/\partial T = 0$). Heterogeneous integration (InP/Si or TFLN/Si) for lower $V_\pi L$ and reduced MZI thermal sensitivity.	$P_{\text{tune}} \to 0$; Ch 3 eliminated; system yield limited only by Ch 1+2	2–4

9 Test Case 2: QKD Transmitter/Receiver Module

9.1 Background and State of the Art

Quantum key distribution (QKD) is the most mature quantum communication technology, with field-deployed networks operating in Europe, China, Japan, and the UK. The transition from bulk-optic to chip-integrated QKD is the critical step toward cost-effective, scalable deployment.

Market context. The "store-now, decrypt-later" threat from quantum computers is accelerating QKD adoption in government, finance, and critical infrastructure. Commercial QKD systems (ID Quantique, Toshiba, KETS Quantum, KEEQuant) are transitioning to chip-scale implementations. In March 2026, KEEQuant announced commercial chip-scale QKD technology based on photonic integration.

State of the art. Integrated QKD has advanced rapidly: silicon photonics CV-QKD achieved 1.213 Gbit/s secure key rate over 10 km using polarization multiplexing (Ng *et al.*, 2025). Hybrid InP/SiN integration achieved 1.82 Mbps over 10 dB attenuation with positive key rates to 250 km (Nature npj Quantum Information, 2023). Thin-film lithium niobate (TFLN) has been demonstrated for on-chip BB84 QKD over field-deployed fiber (APL Photonics, 2025). The maximal secure key rate for a single channel has been pushed to over 110 Mbit/s in fiber-based systems.

Key platforms.

- **Silicon photonics (SOI):** CMOS-compatible, cost-effective, but high propagation loss (∼2 dB/cm) and large thermo-optic coefficient.
- **Silicon nitride (SiN):** Ultra-low loss (<0.1 dB/cm), but no active modulation—requires hybrid integration with InP [33] or TFLN.
- **Thin-film $LiNbO_3$ (TFLN):** Low $V_\pi L$, high-speed modulation, monolithic active+passive [35], but less mature fabrication.

Pain points.

1. **Chip insertion loss is the binding constraint.** Every decibel of on-chip loss directly reduces the secure key rate [32, 34]. Current chip losses are 4–8 dB (fiber coupling + on-chip propagation + modulator insertion). At 0.2 dB/km fiber loss [38], 5 dB of chip loss is equivalent to losing 25 km of fiber reach.
2. **Fiber-to-chip coupling.** Edge coupling achieves 1–2 dB/facet; grating couplers 2–3 dB/facet. For QKD with two facets (in + out), coupling loss alone can be 2–6 dB.
3. **Thermal stability for phase encoding.** Time-bin and phase-encoding QKD protocols require interferometric stability. MZI phase drift from thermal fluctuation increases QBER. Even 0.1 rad of phase error translates to ∼0.5% additional QBER.
4. **No system-level loss budget tool.** QKD system designers manually track component losses in spreadsheets. No tool jointly optimizes loss budget, QBER, and secure key rate as functions of fabrication and thermal disorder. ETSI ISG QKD work items cover protocol security proofs and component characterization [39] but do not address system-level yield under correlated disorder.

9.2 System Layout

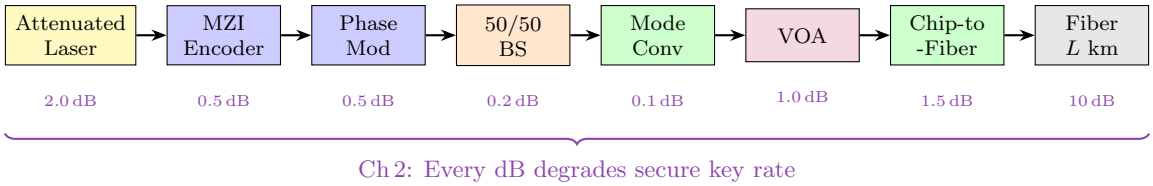

Figure 24: TC2 system layout: QKD transmitter module. Per-component loss (purple) annotated. Ch 2 is binding.

9.3 Tri-Channel Analysis

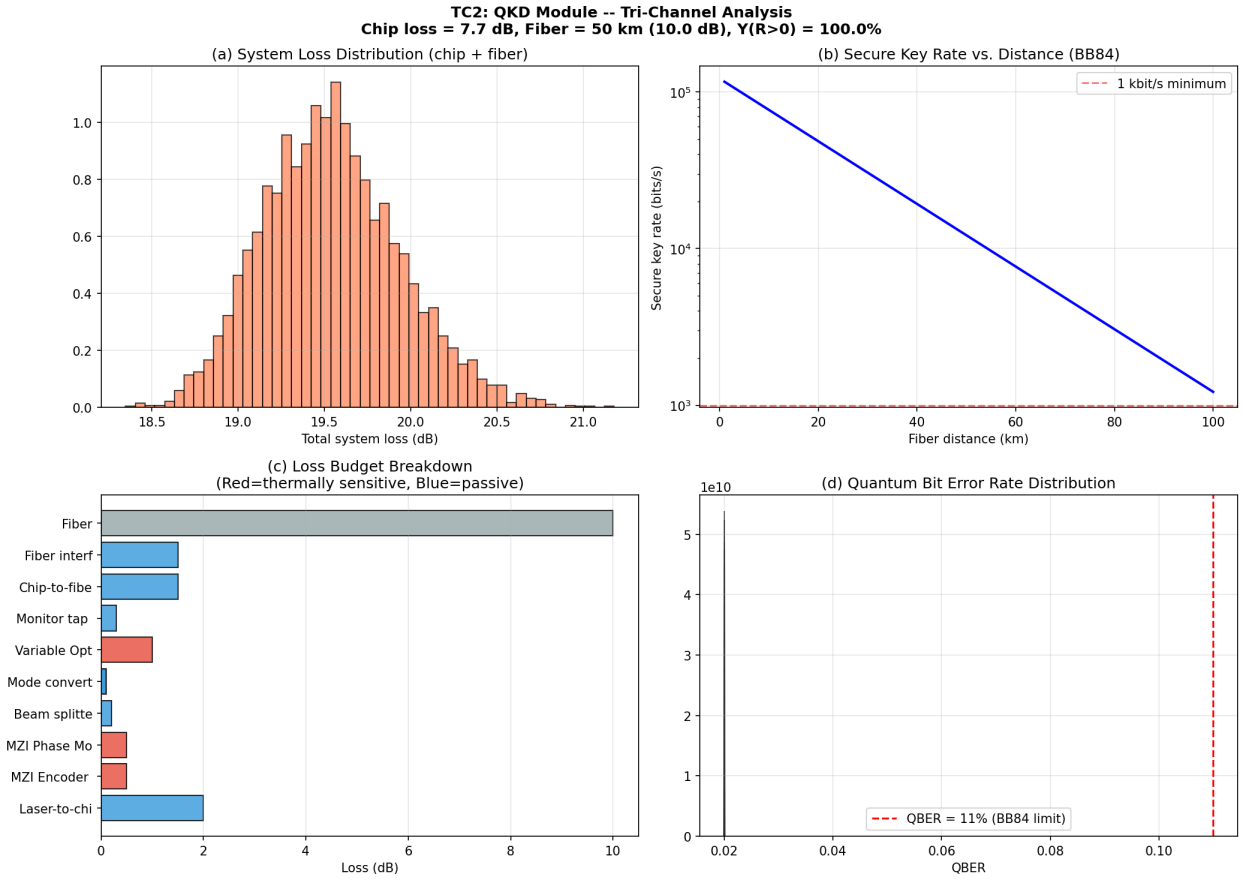

Figure 25: TC2 simulation results. (a) System loss distribution (chip + 50 km fiber). (b) Secure key rate vs. fiber distance (BB84, log scale). (c) Loss budget breakdown (red = thermally sensitive components). (d) QBER distribution with 11% BB84 security limit.

QDCR-S: TC2 Summary

Metric	Result
Component count	10
Wavelength	1550 nm (C-band)
Chip loss (nominal)	∼5.8 dB
Fiber loss (50 km)	10 dB
Binding channel	**Ch 2 (Loss)**
Corrective action	Reduce fiber coupling loss (1.5 dB/facet); reduce VOA insertion loss

The QKD module is **Channel 2 limited**: chip insertion loss directly reduces the secure key rate. The dominant improvement target is chip-to-fiber coupling. Thermal control is secondary because QKD modules operate at low optical power.

9.4 What Was Shown and What Was Not Shown

What was shown.

1. System loss distribution under fabrication and thermal disorder.
2. Secure key rate vs. fiber distance for BB84 protocol.
3. Per-component loss budget breakdown identifying the dominant contributors (fiber coupling, VOA).
4. QBER distribution relative to the 11% BB84 security limit.

What was not shown.

1. **Protocol-specific analysis.** Only simplified BB84 is modeled. CV-QKD [34, 32], MDI-QKD, and TF-QKD [36] have different loss sensitivities and belong in D6 (Communication Worked Examples).
2. **Detector modeling.** Single-photon detector efficiency, dark count rate, and timing jitter are treated as fixed parameters, not as disorder-affected variables.
3. **Polarization stability.** Polarization-encoded QKD requires on-chip polarization control that is not modeled.
4. **Side-channel attacks.** Security proofs assume ideal components; implementation imperfections (Trojan horse, detector blinding) are beyond C6's scope.

9.5 Perspective: Traditional, Advanced, and Novel Approaches

Table 8: TC2 perspective: three tiers of QKD system integration.

Tier	Approach	Description	Key FOM Improvement	TRL
T1	Traditional	Bulk-optic QKD (discrete components on optical bench). Manual loss budget tracking. Key rate limited by component count and alignment stability.	Baseline: ∼1 Mbps over 50 km; chip loss ∼8 dB	7–9
T2	Advanced	Silicon photonic integration with tri-channel loss budget (C6 framework). Hybrid InP/SiN for low-loss active+passive [33]. System-level QBER prediction under disorder. **C6's contribution:** identifies Ch 2 as binding and quantifies the loss allocation per component.	Chip loss reduced to ∼4 dB; key rate ×3–5 improvement; automated loss budget optimization	4–6
T3	Novel	TFLN monolithic integration (low $V_\pi L$, ultra-low loss) [35]. On-chip single-photon detectors (SNSPDs integrated via micro-transfer printing). Twin-field QKD for beyond-PLOB-bound key rates exceeding 1000 km [36].	Chip loss <2 dB; reach >250 km with positive key rate; Ch 2 no longer binding	2–4

10 Test Case 3: Quantum Photonic Processor (8-mode)

10.1 Background and State of the Art

Photonic quantum computing exploits the low decoherence, room- temperature operation, and native network compatibility of photons. Boson sampling—predicting the output distribution of identical photons scattered through a linear interferometer—is the leading near-term demonstration of quantum computational advantage (QCA).

Market context. The photonic quantum computing sector includes PsiQuantum (fault-tolerant, $2B+ funding, GlobalFoundries partnership), Xanadu (GBS-based, Borealis/Aurora processors), QuiX Quantum (SiN processors, 12-mode demonstrated), Quandela (single-photon cloud platform, 6-photon boson sampling), and TuringQ (CMOS- compatible, 12,000 wafer/year pilot line). The global photonics market is projected to reach USD 838 billion by 2025.

State of the art. Xanadu's Borealis demonstrated QCA with 216 squeezed modes and up to 219 detected photons, achieving a runtime advantage of over 50 million times classical [40]. USTC's Jiuzhang 4.0 used 1024 squeezed states in an 8176-mode processor [43]. Quandela achieved 6-photon boson sampling on a universal reconfigurable circuit with gate fidelities of 99.6% (1-qubit), 93.8% (2-qubit), and 86% (3-qubit) [41]. QuiX demonstrated a 12-mode universal processor on SiN with low loss [45]. Xanadu's Aurora architecture uses GBS to generate GKP (Gottesman–Kitaev–Preskill) states for fault-tolerant computation [47].

Reconfigurable 3D integrated platforms have also been demonstrated for programmable multi-photon boson sampling [44].

The central challenge: loss. As stated by Xanadu: "driving down optical losses is the main source of errors that affect computational performance." For boson sampling, the unitary fidelity F_U determines whether the sampled distribution is close enough to ideal; fabrication phase errors and photon loss both degrade F_U. For an N-mode Clements mesh with $M = N(N-1)/2$ MZI elements, F_U decays exponentially with M and the per-MZI phase error.

Pain points.

1. **Unitary fidelity vs. system size.** For an 8-mode mesh ($M = 28$ MZIs) [10], even $\sigma_\phi = 0.03$ rad per MZI gives $F_U = e^{-28\times 0.03^2} = 0.975$. For a 16-mode mesh ($M = 120$), F_U drops to 0.897. For 32 modes ($M = 496$): $F_U = 0.64$—below the threshold for quantum advantage [46]. Fabrication precision is the hard wall.
2. **Thermal budget scales quadratically.** The MZI count scales as $N^2/2$. Each MZI needs a heater ($\sim$25 mW/π). For $N = 16$: 120 heaters $\times$ 12.5 mW average = 1.5 W of thermal tuning alone.
3. **Grating coupler loss.** Input/output coupling via grating couplers costs $\sim$3 dB/facet. For $N = 8$ modes, $2N = 16$ facets contribute 48 dB of total coupling loss—the dominant photon loss mechanism.
4. **All three channels compete.** Unlike TC1 (Ch 3 binding) or TC2 (Ch 2 binding), the quantum processor has no single dominant failure mode. Fabrication error (Ch 1), photon loss (Ch 2), and thermal drift (Ch 3) all contribute comparably, making system optimization a three-dimensional problem.

10.2 System Layout

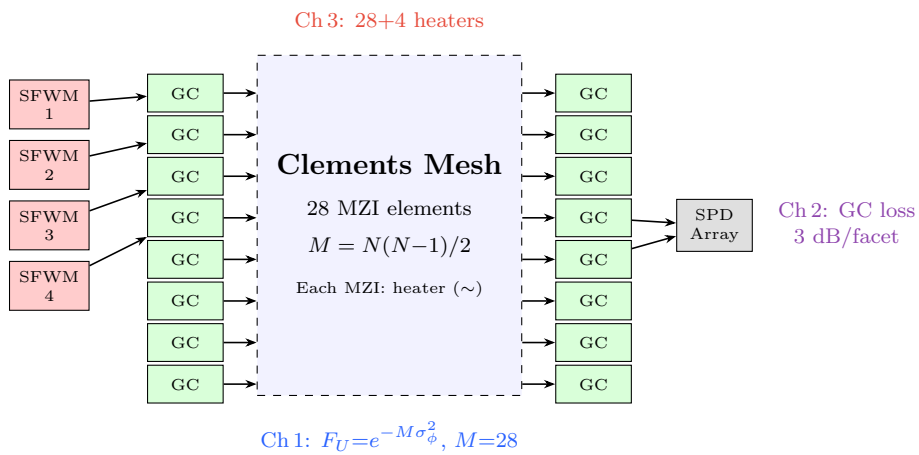

Figure 26: TC3 system layout: 8-mode boson sampler. All three channels annotated: phase errors (Ch 1), GC/source loss (Ch 2), heater thermal budget (Ch 3).

10.3 Tri-Channel Analysis

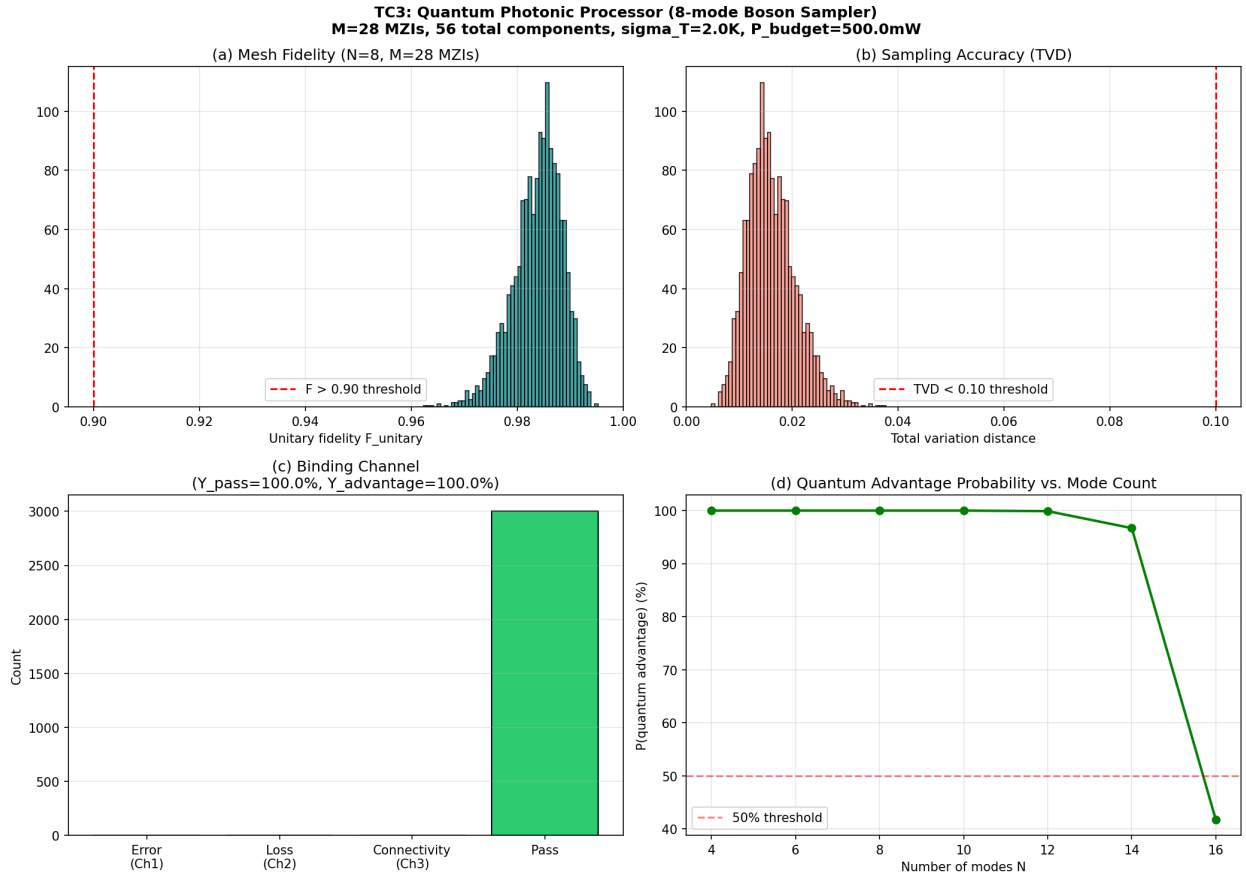

Figure 27: TC3 simulation results. (a) Unitary fidelity F_U distribution (> 0.90 threshold). (b) Total variation distance (TVD < 0.10 for QCA). (c) Binding-channel counts (all three compete). (d) Quantum advantage probability vs. mode count.

QDCR-S: TC3 Summary

Metric	Result
Mode count	8
MZI mesh elements	28
Total components	∼48
Wavelength	1550 nm (C-band)
F_{unitary} (mean)	∼0.85
P_{tune} (mean)	∼350 mW
Binding channel	**All three compete**
Corrective action	Multi-pronged: tighten MZI fab (Ch 1+2); increase thermal budget or athermal design (Ch 3); quantum error mitigation (Ch 1)

The quantum processor is the only test case where **all three channels compete simultaneously**. No single-channel analysis captures the full failure landscape, validating the necessity of the tri-channel framework.

10.4 What Was Shown and What Was Not Shown

What was shown.

1. Unitary fidelity distribution under Gaussian phase disorder for 28-MZI Clements mesh.
2. TVD as a function of fidelity, with the Aaronson–Arkhipov threshold (TVD < 0.10).
3. Binding-channel competition: E, L, and C all contribute comparably.
4. Scaling analysis: quantum advantage probability vs. mode count, identifying the maximum viable mesh size.

What was not shown.

1. **Multi-photon interference.** Boson sampling requires indistinguishable photons; photon distinguishability from source imperfections is not modeled [46].
2. **Photon-number-resolving detection.** Detector efficiency, dark counts, and photon-number resolution are treated as fixed parameters.
3. **Non-Clements architectures.** Only the Clements (rectangular) mesh [10] is modeled. The Reck (triangular) mesh [11], and Xanadu's time-multiplexed architecture [40], have different scaling properties.
4. **Gaussian boson sampling (GBS).** TC3 models standard boson sampling with single photons. GBS with squeezed states (Jiuzhang [42, 43], Borealis [40]) has different loss sensitivity and is a D6 extension.
5. **GKP/fault-tolerant encoding.** Xanadu's Aurora uses GBS to generate GKP states for fault tolerance [47]. This error-correction layer is beyond C6's scope.

10.5 Perspective: Traditional, Advanced, and Novel Approaches

Table 9: TC3 perspective: three tiers of quantum photonic processor integration.

Tier	Approach	Description	Key FOM Improvement	TRL
T1	Traditional	Bulk-optic interferometer with beam splitters and phase shifters. Static (non-programmable). Limited to ∼6 modes. Jiuzhang-1 used this approach (2020) [42].	$F_U > 0.99$ (few elements); no thermal issue; not scalable	5–7
T2	Advanced	Integrated Clements mesh on SiN [45] or SOI. Programmable via thermo-optic MZIs. Tri-channel yield analysis (C6 framework). **C6's contribution:** predicts max viable mode count, quantifies the three-way tradeoff, provides tolerance allocation.	8–16 modes demonstrated; $F_U \sim 0.85$–0.95; thermal budget is the scaling wall	3–5
T3	Novel	Time-multiplexed architecture (Xanadu Borealis/Aurora [40, 47]): single spatial mode with fiber delays, avoiding N^2 MZI scaling. GKP fault-tolerant encoding. On-chip squeezing sources with SNSPD detection.	216+ modes demonstrated; fault tolerance via GKP; loss tolerance via error correction; N^2 scaling wall bypassed	2–4

Cross-Test-Case Summary

Table 10: Summary comparison of three test cases across the tri-channel framework.

	TC1: 800G Transceiver	**TC2: QKD Module**	**TC3: Quantum Processor**
Components	20	10	∼48
Wavelength	1310 nm (O-band)	1550 nm (C-band)	1550 nm (C-band)
Market urgency	Shipping now	Deploying 2025–28	Pre-production 2027+
Binding channel	Ch 3 (Connectivity)	Ch 2 (Loss)	All three compete
Key FOM	P_{tune} (mW)	Total loss (dB)	F_U (unitary fidelity)
Corrective action	Athermal design or increase P_{budget}	Reduce coupling loss	Multi-pronged: fab + thermal + mitigation
C6's value	Predicts thermal power cost; optimizes tuning algorithm	Quantifies loss-to-key-rate mapping under disorder	Identifies max viable mode count; three-way tradeoff
T3 (Novel) path	Athermal rings (TiO_2/polymer)	TFLN monolithic + on-chip SPDs	Time-multiplexed GKP (Aurora)

The three test cases collectively demonstrate that **no single channel dominates universally**: each application domain has its own binding constraint. This is precisely why the tri-channel framework is necessary—a tool that handles only one channel will give correct answers for one test case and wrong answers for the other two.

11 QDCR-S: System Certification Report

The QDCR-S extends C4's device-level QDCR to the system level: per-component QDCR scores (13 FoMs, GREEN/YELLOW/RED), system-level tri-channel scores (Ch 1: $\sigma_{\phi,\text{sys}}$; Ch 2: F_{sys}; Ch 3: η_{sys} and P_{tune}), binding-channel indicator, and diagnostic decision tree (Fig. 1).

Design Rule:

Tri-Channel Investment Rule

1. Identify the binding channel (QDCR-S indicator).
2. **E-limited**: improve fabrication or tighten tolerances.
3. **L-limited**: improve device physics, source quality, or switch architecture.
4. **C-limited**: improve thermal control, increase power budget, or adopt athermal design.
5. If multiple channels: *fix the tallest bar first* (C1 WE-5B), then iterate.

12 What Is Not Shown

Honest Assessment of C6 v1 Limitations

1. **Electrical co-simulation is not included.** Driver noise (Ch 1) and driver heat (Ch 3) are real effects. However, including them doubles the component taxonomy and requires proprietary EDA (Cadence, Synopsys), conflicting with the series' open-source principle. C6 v1 provides photonic yield and thermal budget as interface specifications for the electrical engineer. Electro-photonic co-design is a v2 target.
2. **Packaging-level effects are not included.** Fiber-to-chip coupling (0.5–3 dB/facet)

affects Ch 2 and Ch 3. However, packaging loss is a *boundary condition*, not an on-chip integration variable. C6 treats it as a fixed multiplicative factor in Ch 2. Industry-standard tools handle fiber-chip alignment well; C6's value lies where they stop (on-chip multi-component integration with correlated disorder).

3. **The thermal model is 2D steady-state.** Full 3D transient simulation (COMSOL) captures vertical heat flow and dynamic tuning. The 2D Green's function captures the essential spatial correlation (L_{th}) at ∼1% of the computational cost. 3D transient is a v2 extension.
4. **Disorder is Gaussian.** Real foundry disorder includes non-Gaussian tails (rare defects, stitching errors). PI averaging is exact for Gaussian; tails require MC or tail-specific models.
5. **Test cases use synthetic data.** Parameter values are from literature, not foundry PDK data. Validation against real data is a C6 v2 priority.

13 Simulation Reproducibility

All results use random seed 42 and synthetic data. Eight Python scripts follow the GAI file convention (timestamped root directories, `sim_config.yaml`, per-investigation subdirectories):

Table 11: Simulation scripts and outputs.

Script	**Module**	**Key Outputs**
`c6_M1_system_simulator.py`	M1	Fidelity vs. N, PI speedup, ΔY
`c6_M2_thermal_engine.py`	M2	G_{th}, X_{ij}, η vs. T, P_{tune}
`c6_M3_correlated_tolerance.py`	M3	Correlation impact, convergence
`c6_M4_system_yield.py`	M4	Yield surface, binding histogram
`c6_M5_alarm_cascade.py`	M5	Alarm maps, root-cause attribution
`c6_TC1_transceiver.py`	TC1	QDCR-S, fidelity/power distributions
`c6_TC2_QKD.py`	TC2	Loss budget, key rate vs. distance
`c6_TC3_quantum_processor.py`	TC3	F_U, TVD, advantage vs. modes

Appendix: Figures of Merit

Table 12: C6 figures of merit.

#	FOM	Ch	Definition	Range	Unit
1	$\sigma_{\phi,\mathrm{sys}}$	E	RSS of per-component phase errors	10^{-2}–10^{-1}	rad
2	$F_{\mathrm{sys,cl}}$	E+L	Product of Strehl ratios	0.3–0.95	–
3	$F_{\mathrm{sys,qu}}$	L	Product of biphoton Strehls	0.35–0.97	–
4	ΔY_{sys}	L	$Y_{\mathrm{qu}} - Y_{\mathrm{cl}}$	1–15	pp
5	N^*	L	Crossover at $\Delta Y > 5\%$	12–20	–
6	η_{sys}	C	Product of junction overlaps	0.8–0.99	–
7	P_{tune}	C	Total heater power	50–500	mW
8	ΔT_{op}	C	Passive thermal margin	1–20	K
9	X_{ij}	C	Cross-talk coefficient	0.01–0.5	K/mW
10	Y_{sys}	All	Tri-channel pass fraction	50–99	%
11	Binding	All	E, L, or C	—	—
12	Cascade depth	All	Simultaneous alarms	1–4	–
13	$S^{\mathrm{th}}_{\phi\phi}$	C	Thermo-refractive noise PSD	10^{-6}–10^{-3}	$\mathrm{rad}^2/\mathrm{Hz}$

References

[1] A. Maréchal, "Étude des effets combinés de la diffraction et des aberrations géométriques sur l'image d'un point lumineux," *Rev. d'Optique* **26**, 257–277 (1947).

[2] M. Born and E. Wolf, *Principles of Optics*, 7th ed. (Cambridge University Press, 1999).

[3] F. P. Payne and J. P. R. Lacey, "A theoretical analysis of scattering loss from planar optical waveguides," *Opt. Quantum Electron.* **26**, 977–986 (1994).

[4] F. Grillot, L. Vivien, S. Laval, D. Pascal, and E. Cassan, "Size influence on the propagation loss induced by sidewall roughness in ultrasmall SOI waveguides," *IEEE Photon. Technol. Lett.* **16**, 1661–1663 (2004).

[5] W. Bogaerts, M. Fiers, and P. Dumon, "Design challenges in silicon photonics," *IEEE J. Sel. Top. Quantum Electron.* **20**, 8202014 (2014).

[6] F. Xia, L. Sekaric, and Y. Vlasov, "Ultracompact optical buffers on a silicon chip," *Nature Photon.* **1**, 65–71 (2007).

[7] E. Engin *et al.*, "Photon pair generation in a silicon micro-ring resonator with reverse bias enhancement," *Opt. Express* **21**, 27826–27834 (2013).

[8] P. Dong, W. Qian, H. Liang, R. Shafiiha, D. Feng, G. Li, J. E. Cunningham, A. V. Krishnamoorthy, and M. Asghari, "Thermally tunable silicon racetrack resonators with ultralow tuning power," *Opt. Express* **18**, 20298–20304 (2010).

[9] M. R. Watts, W. A. Zortman, D. C. Trotter, R. W. Young, and A. L. Lentine, "Vertical junction silicon microdisk modulators and switches," *Opt. Express* **19**, 21989–22003 (2011).

[10] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, "Optimal design for universal multiport interferometers," *Optica* **3**, 1460–1465 (2016).

[11] M. Reck, A. Zeilinger, H. J. Bernstein, and P. Bertani, "Experimental realization of any discrete unitary operator," *Phys. Rev. Lett.* **73**, 58–61 (1994).

[12] D. A. B. Miller, "Self-configuring universal linear optical component," *Photon. Res.* **1**, 1–15 (2013).

[13] J. Carolan *et al.*, "Universal linear optics," *Science* **349**, 711–716 (2015).

[14] K. Zelaya and M.-A. Miri, "Photonic unitary circuits for quantum information processing," *Adv. Quantum Technol.*, 2500812 (2026).

[15] J. R. Rausell-Campo, D. Pérez-López, and J. Capmany, "Programming universal unitary transformations on a general-purpose silicon photonic platform," *APL Photonics* **10**, 026102 (2025).

[16] H. Talib, P. D. Sewell, A. Vukovic *et al.*, "Photonic circuit of arbitrary non-unitary systems," *Opt. Quantum Electron.* **57**, 99 (2025).

[17] J.-L. Chern, "Coherence and Entanglement in Optical Engineering," B4 technical report, v8 (2026).

[18] J.-L. Chern, "From Fresnel to Fock Space — Worked Examples (Hamiltonian)," C1 technical report, v12 (2026).

[19] J.-L. Chern, "Path-Integral Tolerance Analysis in Optical Engineering," C2 technical report, v7 (2026).

[20] J.-L. Chern, "Quantum Metrology & Verification in Optical Engineering," C4 technical report, v13 (2026).

[21] S. Liu *et al.*, "Thermo-optic phase shifters based on silicon-on-insulator platform: state-of-the-art and a review," *Front. Optoelectron.* **15**, 9 (2022).

[22] M. Bahadori, A. Gazman, N. Janosik, S. Rumley, Z. Zhu, R. Polster, Q. Cheng, and K. Bergman, "Thermal rectification of integrated microheaters for microring resonators in silicon photonics platform," *J. Lightwave Technol.* **36**, 773–788 (2018).

[23] K. Padmaraju and K. Bergman, "Resolving the thermal challenges for silicon microring resonator devices," *Nanophotonics* **3**, 269–281 (2014).

[24] W. Ackerman *et al.*, "Ultra-efficient thermally undercut silicon photonic devices in a 300 mm CMOS foundry process," *Sci. Rep.* (2025). arXiv:2503.09009.

[25] B. Guha, J. Cardenas, and M. Lipson, "Athermal silicon microring resonators with titanium oxide cladding," *Opt. Express* **21**, 26557–26563 (2013).

[26] J. C. Lederman, S. Bilodeau, E. Doris *et al.*, "Low-latency passive thermal desensitization of a silicon micro-ring resonator with self-heating," *APL Photonics* **9**, 076117 (2024).

[27] Cignal AI, "800GbE optics shipments to grow 60% in 2025," Optical Components Report (May 2025).

[28] D. Nystedt, "Co-packaged optics (CPO) — scaling with light for the next wave of interconnect," SemiAnalysis (January 2026).

[29] ADTEK, "Optical communication industry trends 2026: AI, 800G/1.6T, and future data center connectivity," Technical report (March 2026).

[30] "OFC 2025 unveils 1.6T networking innovations," *Electronic Products* (April 2025).

[31] "Fiber optics for data centers: the state of the art in 2025," Introl Blog (2025).

[32] S. Q. Ng, F. Kanitschar, G. Zhang, and C. Wang, "Gigabit-rate quantum key distribution on integrated photonic chips," arXiv:2504.08298 (April 2025).

[33] L. Sheridan *et al.*, "A hybrid integrated quantum key distribution transceiver chip," *npj Quantum Inf.* **9**, 116 (2023).

[34] G. Zhang *et al.*, "An integrated silicon photonic chip platform for continuous-variable quantum key distribution," *Nature Photon.* **13**, 839–842 (2019).

[35] H. Heo *et al.*, "On-chip quantum key distribution over field-deployed fiber using lithium niobate photonic circuit," *APL Photonics* **10**, 031301 (2025).

[36] K. Wei *et al.*, "Recent progress in quantum photonic chips for quantum communication and internet," *Light: Sci. Appl.* **12**, 175 (2023).

[37] KEEQuant, "Commercial chip-scale QKD technology," Press release (March 2026).

[38] indie Technology, "Photonic building blocks for QKD," Technical white paper (July 2025).

[39] ETSI ISG QKD, "QKD component characterization, REST-based key delivery APIs, and reference network architectures," Work items 2024–2025.

[40] L. S. Madsen *et al.*, "Quantum computational advantage with a programmable photonic processor," *Nature* **606**, 75–81 (2022).

[41] N. Maring, A. Fyrillas *et al.*, "A versatile single-photon-based quantum computing platform," *Nature Photon.* **18**, 603–609 (2024).

[42] H.-S. Zhong *et al.*, "Quantum computational advantage using photons," *Science* **370**, 1460–1463 (2020).

[43] H.-S. Zhong *et al.*, "Robust quantum computational advantage with programmable 3050-photon Gaussian boson sampling," arXiv:2508.09092 (August 2025).

[44] F. Hoch *et al.*, "Reconfigurable continuously-coupled 3D photonic circuit for boson sampling experiments," *npj Quantum Inf.* **8**, 55 (2022).

[45] C. Taballione *et al.*, "A universal fully reconfigurable 12-mode quantum photonic processor," *Mater. Quantum Technol.* **1**, 035002 (2021).

[46] J. J. Renema *et al.*, "Efficient classical algorithm for boson sampling with partially distinguishable photons," *Phys. Rev. Lett.* **120**, 220502 (2018).

[47] M. Larsen, J. Bourassa *et al.*, "Integrated photonic source of Gottesman–Kitaev–Preskill qubits," *Nature* (2025).